

Formalisation Report

On the inductive blockwise Alperin weight condition
for type B and type C

This report explains arguments concerning the inductive blockwise Alperin weight condition for simple groups of Type B, simple groups of Type C and sporadic simple groups, together with their consequence for finite groups. The results checked in Lean depend on the stated assumptions about the groups, characters, blocks and computational data. The exposition identifies the Mathlib foundations, general ModularRep results and deductions for these families.

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Chapter 1

The inductive condition and the scope of the arguments

The blockwise Alperin weight (BAW) conjecture predicts that a block has as many irreducible Brauer characters as conjugacy classes of weights. The inductive blockwise Alperin weight (iBAW) condition requires compatible equivariant correspondences on covering groups of a nonabelian simple group. These include conditions on character extensions and on blocks of intermediate groups.

This report follows the manuscript's arguments for the inductive blockwise Alperin weight condition for simple groups of Type B, simple groups of Type C and sporadic simple groups. It begins with the general preliminary results, then treats the three families and the reduction to finite groups. The conclusions hold under the hypotheses stated in each argument. References to manuscript results use the numbering of the accompanying manuscript. The mathematical library manual describes the foundations and the general ModularRep results used in these arguments.

1.1 Characters, weights and covering groups

Fix a prime ℓ . A weight of a finite group G consists of a radical ℓ -subgroup Q and an ordinary irreducible character θ of $N_G(Q)/Q$ with defect zero. The set $\text{Alp}(b)$ consists of the conjugacy classes of weights belonging to a block b , and $\text{IBr}(b)$ denotes its irreducible Brauer characters. For passage through a quotient or an isomorphism, both the subgroup map and the character map must be specified under the stated hypotheses.

The full universal covering group of a simple group S and its universal ℓ' -cover $X_{\ell'}(S)$ play different roles. Each descent argument uses the specified covering projection, central kernel and automorphism action. The constructions for exceptional covers include these maps and the character and block identifications they induce.

Implementation.

The representations, irreducibility predicate and ordinary character function are supplied by Mathlib. ModularRep uses these definitions to form its type of ordinary irreducible character functions, and provides the Brauer character and block constructions used below. The mathematical library manual explains the imported definitions and results, the additional constructions, and the explicit hypotheses.

1.2 Coefficient fields and reduction

Brauer characters use a chosen correspondence between roots of unity in characteristic ℓ and characteristic zero. Character comparisons use compatible roots.

The coefficient hypotheses depend on the application. The principal 2-block construction for Type C, including its rank two principal case, uses an algebraically closed coefficient field K for ordinary characters. It does not require a discrete valuation ring with fraction field K . The applications in defining characteristic and in low rank over even fields use a field K of characteristic zero, an algebraically closed field k of characteristic ℓ , and the specified correspondences between roots of unity. They do not require K to be algebraically closed. The application of Li–Li’s theorem to all blocks in rank two and the applications in higher rank at odd nondefining primes use a specified modular system with fraction field K and residue field k , together with the stated splitting and root assumptions. Fixed embeddings of a common cyclotomic field compare complex character values with values in K .

The hypotheses on the decomposition homomorphism, its naturality and block induction refer to these coefficient choices. Where a published statement uses complex characters, its application over another value field includes the required character and root identifications.

1.3 The conclusions and their assumptions

The equality

$$|\mathrm{IBr}(b)| = |\mathrm{Alp}(b)|,$$

an equivariant bijection between these sets, and a bijection satisfying the prescribed block isomorphism relation between modular character triples are distinct conclusions. The full inductive condition also requires compatible extensions and block equalities for intermediate groups. At the trivial radical subgroup it prescribes the correspondence coming from ordinary characters of defect zero.

The arguments below state which conclusion is obtained. Counting arguments give numerical equalities or equivariant bijections under the stated group actions. Published criteria supply the further character and block conditions under their hypotheses. When a passage from individual blocks to a full family chooses compatible correspondences, the resulting family need not coincide with independently chosen block correspondences.

Each Lean theorem proves its conclusion under its stated hypotheses. These include standard representation theory, cited results, finite data and their interpretation for the groups in question. The geometric constructions and trace identities used in manuscript Lemmas 2.11 and 2.12 remain assumptions. In particular, the construction of the centrally normalised split pair in Type C is not formalised. Lean proves the finite spanning, independence, coverage and character selection deductions. The conditional Type C selection theorem concerns the specified family in the conformal symplectic group. Induction and restriction identities give a separate finite selection result for the symplectic group. The principal block application takes the published field invariance and character counts as explicit assumptions. The passage from the selected ordinary characters to these principal block conclusions is not formalised. At rank two, Li–Li’s theorem is applied on the specified cover. Compatibility across blocks and the quotient and root identifications required for the central lift remain separate assumptions. The inductive condition reported by CTBlocks for the other 22 sporadic groups is also assumed. A numerical table is used with its stated interpretation and completeness assumptions. The implementation paragraphs identify the deductions that are formalised.

1.4 Scope of the hypotheses

The Type C source interface applies the published block theorem only to the specified automorphism stabilisers. The theorem `Main.theorem_1_1` assumes both `Main.ClassicalInputs` and `Sporadic.Theorem.Inputs`. The package does not construct either complete collection of source assumptions. The conditional deductions do not establish the manuscript without the stated external assumptions.

1.5 References and source records

Appendix A relates the numbered manuscript statements to selected Lean declarations and states their roles. Appendix B indexes the manuscript citations and bibliography. Appendix C describes further constructions in the source library and distinguishes auxiliary results from those used in the manuscript deductions. Numbers within Lean declaration names are not manuscript cross-references. The source records specify the version consulted, result location, relevant hypotheses and use in the formalisation. A supporting reference documents the stated result at that location. It is not, by itself, a claim that the cited work first established the result.

1.6 Compilation and verification

All 1,999 project Lean source files and four audit sources were checked with Lean 4.33.1 using `--trust=0`. The distributed Lean sources agree with the compiled sources outside comments. The build reused precompiled external dependencies at the pinned revisions. The recorded external objects and direct imports were checked against their hashes. The full tree of indirect external dependencies was not checked, and the imported proofs were not replayed.

Compilation and verification used approximately 38.1 CPU hours. The peak sampled sum of the processes' resident working sets was 10.40 GiB. Shared pages can be counted more than once, and peaks between samples may be missed.

The package checker authenticates the distributed sources and metadata without rerunning Lean, and the original build logs and compiled objects are not distributed.

Source files for the supplementary exhaustive audit helper, its generator and the generated audit module are not distributed. That audit cannot be repeated using the supplied tooling.

For the audited declarations, the axiom checks restrict dependencies to the permitted standard axioms. They do not check every declaration in the library. The direct conclusion check rejects a leading hypothesis that is definitionally equal to the conclusion. It does not detect a merely logically equivalent hypothesis, a conclusion hidden in a structure field, or joint circularity among hypotheses. These checks do not establish the truth, independence or joint satisfiability of the external assumptions.

Chapter 2

The preliminary arguments

This chapter follows the preliminary arguments in Section 2 of the manuscript. In the finite group arguments, ℓ is prime. Brauer characters take values in a specified field of characteristic zero, using the chosen correspondence between roots of unity in that field and in the field of characteristic ℓ . Whenever a published statement is expressed with complex characters, its application over another value field includes the stated character and root identifications.

2.1 Central characters and tensor stabilisers

Let Z be a central subgroup of G of order prime to ℓ . A character lies over $\nu \in \text{Irr}(Z)$ when its restriction to Z is a positive integral multiple of ν . A weight (Q, θ) lies over ν when the inflation of θ to $N_G(Q)$ does. These are the central sectors of manuscript Definition 2.1.

Over an algebraically closed field k of characteristic ℓ , the primitive idempotents e_ν of kZ are orthogonal central idempotents of kG with sum one. If b is a primitive central idempotent of kG , exactly one be_ν is nonzero, and then $be_\nu = b$. This defines the sector of b . Every simple module in b has central character ν . The ordinary characters in the block have the same central character by their decomposition on ℓ -regular elements. Brauer's definition of block induction preserves it: compare the central functions on the singleton classes of elements of Z [65, Definition 2.1]. Thus the weight classes in b also lie in this sector.

Automorphisms transport the sector of b by their action on Z . Inner automorphisms fix every block. Both statements are proved on the primitive idempotents, with the same right action convention as that used for the characters.

Implementation.

The `CentralSectors` namespace in `ManuscriptIBAW.Preliminaries` provides the following results from `ModularRep`: `existsUnique_centralCharacterSector`, `primitiveCentralIdempotentSector_eq_centralCharacter` and `rightMulAction_smul_centralCharacterSector`. `ModularRep` constructs the multiplicative central character from the scalar endomorphism statement of Schur's lemma in `Mathlib`. The central sector deductions use these characters and the block definitions in `ModularRep` together with `Mathlib`'s group algebra and idempotent definitions. The application to ordinary characters and weights additionally uses the specified reduction and local laws of block induction.

For manuscript Lemma 2.2, let $N \triangleleft G$ and let C be the group of linear characters of G/N of order prime to ℓ , acting on blocks by tensoring. If $\lambda \in C_b$, the common central character of the Brauer characters of b gives $\lambda|_{Z(G)'} = 1$. Its order makes it trivial on $O_\ell(Z(G))$ as well. Hence it factors through $G/NZ(G)$ and

$$|C_b| \leq |G : NZ(G)|.$$

The same conclusion holds for linear Brauer characters when reduction identifies them with these ordinary characters and commutes with tensoring. The library manual proves the result under these coefficient and tensor compatibility hypotheses. Its additional semidirect product argument is used later to show that the relevant block stabilisers are 2-hypoelementary.

2.2 The full condition and modular character triples

The full inductive condition of Späth requires a partition by radical subgroups, equivariant local bijections, preservation of central characters and induced blocks, and compatible extensions over every required intermediate subgroup [65, Definition 4.1]. For the trivial radical subgroup it also prescribes $\chi^0 \mapsto \chi$ for each ordinary character of defect zero, with identical global and local extensions.

Manuscript Theorem 2.3 restates Späth's character triple formulation [66, Theorem 4.4] of the extension and block compatibility requirements in [65, Definition 4.1(iii)]. It begins on the full universal cover $\widehat{G}$ of a nonabelian simple group. Fix a radical subgroup $\widehat{Q}$, a Brauer character $\widehat{\theta}$ of $\widehat{G}$, and an ordinary character of $N_{\widehat{G}}(\widehat{Q})/\widehat{Q}$ with defect zero whose inflation $\widehat{\theta}'$ belongs to a block inducing to $\text{bl}(\widehat{\theta})$. Let $\widehat{Z}$ be the kernel of the central character of $\widehat{\theta}$ and set

$$G = \widehat{G}/\widehat{Z}, \quad Q = \widehat{Q}\widehat{Z}/\widehat{Z}, \quad M = N_G(Q).$$

Both characters descend to these quotients. The ℓ -part of $\widehat{Z}$ lies in $\widehat{Q}$. Descent through its part of order prime to ℓ uses the equality of the induced blocks.

These extension and block compatibility requirements are equivalent to the required block isomorphism of the modular character triples on G and M . In the extension formulation, there is a group A with $G \triangleleft A$, $C_A(G) = Z(A)$ and $\ell \nmid |Z(A)|$, whose conjugation image is the required character stabiliser. The global and local Brauer characters extend to A and $N_A(Q)$, and the extensions satisfy

$$\text{bl}(\widetilde{\theta}|_J) = \text{bl}(\widetilde{\theta}'|_{N_J(Q)})^J \quad (G \leq J \leq A).$$

The induction superscript J is essential. It is omitted in the printed equation [66, (12)]. The local triple uses the full stabiliser of the weight representative, which must lie in the stabiliser of the global character.

Implementation.

The `SpathCriterion` namespace in `ManuscriptIBAW.Preliminaries` provides `current_theorem_2_3` and its two implications from `ModularRep.FullCover`, `PairData` and `HattedPairJoins` specify the full cover, central quotient, normaliser and the same ordinary local character. `Theorem44Source` supplies the published equivalence uniformly for those data. The modular block triple relation is the parameter `BlockTripleSourceSemantics.blockIsomorphic`. `Theorem44Source` assumes that, under its cover, coefficient and root hypotheses, `ClauseIII` is equivalent to `TripleCondition`. The latter includes containment of the full local stabiliser in the global stabiliser. The source hypothesis therefore constrains the chosen relation. Its identification with the usual definition by projective representations is not formalised, and the transport laws used below remain assumptions. The equivalence concerns the extension and intermediate block conditions. It does not establish the other parts of the inductive condition. Its ordinary field is algebraically closed, with no discrete valuation ring required. This named theorem is a standalone formulation. The main applications use their stated extension and block conditions and the separate criterion sources attached to those applications. They do not invoke `current_theorem_2_3`. In particular, `SpathProposition46Theorem44Source` in `ModularRep.PaperProofs.EvenFieldFLZBAWGoodFamily` assumes the passage from a specified full family on the covering group to equivariant block bijections satisfying the relation specified by

`SpathProposition46Theorem44FamilySemantics`, under its cover and automorphism hypotheses. This assumption constrains the relation on the corresponding pairs. Its interpretation as block isomorphism of modular character triples remains external. The separate `SpathProposition46Theorem44ToDefinition35FamilySource` assumes the implication to the specified Definition 3.5 relation. These application sources are supplied separately from the standalone theorem.

The paragraph following manuscript Theorem 2.3 applies the theorem to a block bijection supplied by the full inductive condition. After inflation to the full universal cover, the theorem gives the character triple condition on the quotient by the kernel of the central character. Inflation of this relation gives the condition for an iBAW bijection [19, Remark 4.4]. This uses transport of a character triple relation through a central kernel. The general transport law is an external assumption in the formalisation. It is distinct from the central ℓ -group argument for manuscript Lemma 2.6 below.

2.3 Combining blocks and descending to the simple group

Manuscript Remark 2.4(a) combines Koshitani–Späth’s Lemma 3.3 [50] and Späth’s Definition 5.17 and Remark 5.18 [65]. The full inductive BAW condition for one block in each automorphism orbit gives the relative condition, and the relative conditions combine over the required defect groups. Taking all blocks and all radical subgroups gives the full condition for the simple group.

Part (b) of the remark gives an elementary observation about finite group actions. Suppose that A permutes disjoint families of sets $\mathcal{U}_b$ and $\mathcal{V}_b$ through its action on their index set. Bijections $\Phi_b : \mathcal{U}_b \rightarrow \mathcal{V}_b$ that commute with A_b , chosen for representatives of the A -orbits, extend to one A -equivariant bijection by $\Phi(x^a) = \Phi_b(x)^a$. This observation applies to tensor actions as well as automorphism actions. It combines bijections and does not supply the additional character and block conditions in the full inductive condition.

Implementation.

The `BlockFamilies` namespace in `ManuscriptIBAW.Preliminaries` provides the `ModularRep` application in which all blocks are included. Its `Source` assumes the published passage from the blockwise conditions to the full condition, including its normalisation, over one specified modular system. The blocks, local block operations and root correspondence refer to that system. When the full inductive condition holds for every block, `fullFamily` applies that hypothesis to obtain a compatible family for all blocks. The source includes the published passage to common choices, which can change the individual correspondences. The reduction for the full condition in Remark 2.4(a) is used through this special case. This provider does not formalise the general construction in part (b). The ordinary coefficient field is the fraction field of the specified discrete valuation ring, with the required roots. It need not be algebraically closed.

For manuscript Corollary 2.5, suppose that the simple group S satisfies the full inductive condition at ℓ . Späth’s Proposition 4.6, with the perfect subgroup equal to S , group $\text{Aut}(S)$ and multiplicity one, supplies the partition, block bijections and extension information [65, Proposition 4.6(a)–(c)]. The conversion to modular character triples uses [66, Proposition 3.6(b) and Theorem 3.5(b)]. It gives an iBAW bijection on every block of S .

Implementation.

The `Descent` namespace in `ManuscriptIBAW.Preliminaries` provides the descent construction from `ModularRep`. Its `FullCoverPrimeToDiagram` fixes the full cover, its quotient by the central ℓ -core, and the simple group. The Brauer inflation, ordinary local characters, blocks and coefficient comparisons refer to these maps. Lean constructs the required epimorphism from the ℓ' -cover to the target. `Proposition46Theorem44Source` applies the published descent uniformly to that epimorphism and the assumed full inductive condition. `current_corollary_2_5_coherent` gives

the coherent family, and `current_corollary_2_5` identifies its triples with those in the definition of an iBAW bijection. These interfaces use algebraically closed ordinary fields and no modular system. Their covering maps, root comparisons and published implications are distinct assumptions. These named algebraically closed formulations are standalone results. The exceptional Type B application uses the finite splitting formulation in `ModularRep.CurrentSpathFiniteSplittingDescent`. Its theorem `current_corollary_2_5_block` transports the bijection of an established covering family through the stated character and weight correspondences. The external source supplies the character conditions for those same pairs, under explicit splitting, root and automorphism stabiliser hypotheses.

2.4 Lifting a bijection through the central core

For manuscript Lemma 2.6, let $G \triangleleft A$ and suppose that $P = O_\ell(G) \leq Z(G)$. Write $\overline{G} = G/P$ and $\overline{A} = A/P$. Every radical ℓ -subgroup Q of G contains P , and we write $\overline{Q} = Q/P$. Let b be an A -stable block and let $\overline{b}$ be its dominated block on $\overline{G}$. Suppose that $\overline{b}$ has an iBAW bijection compatible with $\overline{A}$.

Inflation identifies the Brauer characters of these two blocks (see, for example, [58, Theorem 9.10]). Centrality of P supplies the hypothesis of this block correspondence. The quotient map induces

$$N_G(Q)/Q \cong N_{\overline{G}}(\overline{Q})/\overline{Q}.$$

Transport the weight character along this isomorphism. With the block correspondence, this gives the weight bijection [30, Lemma 2.1]. Both identifications commute with the action of A . Composing them with the given bijection therefore gives an A -equivariant bijection on b .

Let (Q, θ) represent the weight class corresponding to χ . The same subgroup P lies in $N_G(Q)$ and in the kernels of χ and the inflated Brauer reduction θ^0 . The quotient identifies the character and weight stabilisers with their counterparts in $\overline{A}$. The implication for a normal kernel in Martínez–Rizo–Rossi therefore lifts the block isomorphism of the associated triples [57, Lemma 3.14]. This proves compatibility with A for every representative of the corresponding weight class.

Implementation.

The `CentralCore` namespace in `ManuscriptIBAW.Preliminaries` provides the following quotient results from `ModularRep`: `rawLocalQuotientEquiv`, `quotientRawWeight` and `ownReduction_inflation`. The last identity compares reductions of the same ordinary local character. `lemma_2_6_compatibleBijection` constructs the bijection on the specified block from the bijection on the quotient block, proves equivariance, and applies `sameWeight_relation_upward`. Equivariance and injectivity also show that the stabiliser of the weight representative is contained in the stabiliser of its Brauer character.

`CentralBlockSource` specifies the centrality hypothesis and the block correspondence.

`TripleQuotientSource` identifies the quotient stabilisers, their character values and roots, and the block isomorphism relation in those coordinates. The relation for the lifted characters then follows from the published implication for a normal kernel. `returnCompatibleThroughCentralQuotient` then identifies the constructed family with the intended blocks and definition of an iBAW bijection. It does not require the quotient to be a universal cover. The application over a modular system uses the stated splitting conditions on its fraction field, without requiring algebraic closure.

2.5 Permutation lattices and cyclic extensions

For manuscript Corollary 2.7, let A be r -hypoelementary and let Y, Y' be finite A -sets. Every subgroup of A is again r -hypoelementary. Conlon’s permutation lattice theorem [25], in the formulation of Bouc [12, Theorem 3.5.5], therefore gives $|Y^U| = |Y'^U|$ for every subgroup

$U \leq A$ whenever $\mathbb{Z}_r[Y] \cong \mathbb{Z}_r[Y']$. Burnside's injectivity theorem for the mark homomorphism [12, Theorem 2.3.2] then gives $Y \cong Y'$ as A -sets.

Manuscript Lemma 2.8 applies this to an A -stable integral basic set $\mathcal{X}$ for an ℓ -block b . The decomposition map restricts to an isomorphism

$$d_b : \mathbb{Z}[\mathcal{X}] \longrightarrow \mathbb{Z}[\text{IBr}(b)]$$

[43, (1.1)–(1.2)]. If d_b is equivariant, it is an isomorphism of permutation lattices. Extend scalars to $\mathbb{Z}_2$ and apply the preceding argument when A is 2-hypoelementary. The prime used for the permutation lattice is 2, independently of ℓ . This proves existence of an equivariant bijection. It specifies neither an image for each character nor a unitriangular decomposition matrix. The library manual gives the lattice and mark arguments and the hypotheses of their formal statements.

Implementation.

In `ModularRep.PaperProofs.ConlonBasicSet`, `corollary_2_4_conlonMark` implements manuscript Corollary 2.7. It deduces an equivariant bijection from the permutation lattice isomorphism, assuming `PadicConlonMarkDetection` and `PublishedBurnsideMarkInjectivity`. These hypotheses state the two published results for arbitrary finite A -sets. The same module's `lemma_2_5_basicSetBridge` implements the lattice argument in manuscript Lemma 2.8. Starting from an integral linear equivalence for the restricted decomposition map and its equivariance, it constructs the permutation lattice isomorphism, extends scalars to $\mathbb{Z}_2$, and applies the preceding deduction. The existence and equivariance of the restricted decomposition map remain hypotheses of this declaration.

Manuscript Lemma 2.9 is the cyclic extension theorem. If $N \triangleleft H$ and H/N is cyclic, an invariant irreducible ordinary or Brauer character of N extends to H . For examples of these results, see [47, Corollary 11.22] for ordinary characters and [58, Theorem 8.12] for Brauer characters. The ordinary statement uses complex characters or a compatible splitting interpretation. The modular statement uses algebraic closure of the field of characteristic ℓ and requires no coprimality condition on H/N .

Implementation.

`ModularRep.OrdinaryCharacterCyclicExtensionBridge` proves `exists_extension_of_fixed_cyclic_quotient`, using `Representation.CyclicExtensionPrinciple`. `ModularRep.BrauerCharacterExtensionBridge` proves `exists_extension_realisation_of_ibr_fixed_cyclic_quotient`, using `Representation.BrauerCyclicExtensionPrinciple`. The Brauer character is then constructed in `ModularRep.BrauerCharacterHomPullback` by `Representation.Extension.brauerCharacterExtensionWitnessOfCompatible`, with compatible root correspondences. These constructions prove invariance of the chosen realisations and the character restriction identities. The representation extension principles remain explicit hypotheses. The ordinary principle over the chosen field quantifies over all finite ambient groups. Splitting for the base group alone does not establish it. Isaacs's theorem gives the complex case, and its interpretation over the chosen ordinary field is an additional assumption. The modular principle uses an algebraically closed field. Neither extension principle is proved in these declarations.

Finally, manuscript Lemma 2.10 assumes that the simple group G is its own universal cover and that $\text{Aut}(G) = G \rtimes E$ with E cyclic. For an E -stable block b , an E -equivariant bijection $\text{IBr}(b) \rightarrow \text{Alp}(b)$ extends to equivariance under $G \rtimes E$, since inner automorphisms fix both sets. Each Brauer character extends to its stabiliser by the cyclic extension theorem. For a weight (Q, θ) , set $D = (G \rtimes E)_{Q, \theta}$. The quotient $D/N_G(Q)$ embeds in E . Apply the same extension theorem to the reduction of the original local character on $N_G(Q)/Q$ inside D/Q , then inflate to D .

These extensions and the block equality attached to the weight verify the specialised Brough–Späth criterion with $\tilde{G} = G$ [31, Theorem 3.18]. The conclusion is that b is BAW-good. Applications use the specified cover, local character, block induction and extension interpretations. A full family conclusion also uses the separate passage from the block conditions to compatible correspondences.

Implementation.

In `ModularRep.PaperProofs.CurrentCyclicOuterBAW`, `lemma_2_10_bawGood` applies the published criterion, supplied as `FLZ318BAWGoodSource`, to the hypotheses constructed by `completedExplicitPackage_of_cyclicEndgame`. Its conclusion is BAW-goodness of the specified block. The extension principles, compatible roots, structural and block assumptions, and identifications of the covering group and character triple relation remain explicit. This declaration does not prove the published criterion or the separate passage to a compatible family for all blocks.

2.6 Character selection for finite reductive groups

The manuscript proves two character selection lemmas for connected reductive groups. Lemma 2.11 concerns isolated families in Type B. Lemma 2.12 concerns selected Type C families in rank at least three. Their proofs are in Appendix B of the manuscript. For the latter, the semisimple parameter is chosen in a split torus and multiplied by a central element to obtain a parameter whose square is one. The family in the conformal symplectic group is attached to this rational pair. Both lemmas select ordinary characters whose positive Alvis–Curtis duals have Kronecker delta scalar products with the generalised Gelfand–Graev characters of the rational classes in the chosen geometric unipotent class. The arguments and their source assumptions are discussed in Chapters 4 and 3.

Implementation.

The geometric constructions, including the central choice in Lemma 2.12, are not formalised. The character theoretic identities used in the finite proofs are explicit assumptions. The Type B application proves the finite spanning, independence, coverage and selection deductions from those identities. In `ManuscriptIBAW.TypeC.GGGRSelection`, `delta_selection` proves a conditional form of Lemma 2.12 on the specified conformal symplectic group. From the stated component action, weighted formula, denominator equality and trace identities, the proof shows that the component indices are one and the scalar product row sums equal one. It then constructs an injective choice of characters. The selected family and its geometric interpretation are supplied separately. The general finite arguments are described in the library manual’s chapter on character counts from finite matrices.

Chapter 3

The Type C arguments

Let F be a finite field of order $q = p^f$, let $n \geq 2$, and set $G = \mathrm{Sp}_{2n}(F)$ and $S = \mathrm{PSp}_{2n}(F)$. The pair $(n, q) = (2, 2)$ is excluded. The argument separates the defining characteristic from the nondefining primes, then treats the parity of q and the small ranks.

For $\ell = 2$ with q odd, principal weights are parametrised by their subgroup factors and irreducible characters of the corresponding normaliser quotients. Over an even field, the correspondence between unipotent characters and generic weights treats the unipotent blocks needed for Jordan reduction. For odd nondefining primes over odd fields, an integral basic set and Conlon's theorem give the bijection between Brauer characters and ordinary characters needed for the stabiliser argument. Jordan reduction uses a suitable automorphism group for each semisimple parameter.

Throughout, the hypotheses identify the finite groups, their blocks and characters, and the common roots used for ordinary and Brauer character values. The distinctions between these interpretations and the proved deductions are set out in Chapter 1.

3.1 The principal block in odd characteristic

At rank two, manuscript Proposition 3.3 uses [53, Theorem 1.1]. For odd q_0 , this gives the inductive condition for $\mathrm{PSp}_4(q_0)$ at 2. Manuscript Corollary 2.5 gives an iBAW bijection on its principal block. Manuscript Lemma 2.6 lifts this bijection to the principal block of $\mathrm{Sp}_4(q_0)$ through its central subgroup of order two. This supplies the rank two principal block required by Jordan reduction. The construction below applies in rank at least three.

We use the principal weight criterion in [36, Lemma 2.3]. For a 2-subgroup R and an irreducible character φ of $N_G(R)$, the criterion concerns the same character in three conditions: $Z(R)$ is a Sylow 2-subgroup of $C_G(R)$, the subgroup $RC_G(R)$ is contained in $\ker \varphi$, and the character on $N_G(R)/RC_G(R)$ has defect zero.

Radicality follows from the local character conditions. A quotient with an irreducible ordinary character of defect zero has no nontrivial normal 2-subgroup. For the constructed subgroup, $C_G(R) \leq R$, so $RC_G(R) = R$ and $C_G(R) = Z(R)$. The same quotient character then satisfies the principal criterion. This proves that the constructed pair is a principal weight.

Proposition 3.48 of [36] lists the possible basic factors. The manuscript excludes case (4) by a reflection argument and uses Remark 3.50 for cases (1), (2) and (5). The formalisation assumes these three exclusions and derives the exclusion of case (4) from the reflection argument. For each irreducible character of the relevant normaliser quotient, its multiplicity is the size of a staircase partition. Each such multiplicity is a triangular number, but their

sum need not be a triangular number. Manuscript Remark 3.4 illustrates this distinction with $Q_8 \times Q_8$ in $\mathrm{Sp}_4(3)$. The Lean results `twoColouredHeights_size` and `triangular_ne_two` in `ModularRep.PaperProofs.OddTwoWreathCoreCharacterSource` prove the arithmetic example with heights $(1, 1, 0)$. They do not identify its symplectic subgroup or normaliser quotient.

On each symplectic summand, the centraliser calculation is the applicable part of An's argument [2, p. 198]. The independent scalar transformations which act as $-I$ on one summand and as I on the others belong to the constructed radical subgroup. Commutation with these signs forces each matrix entry between distinct summands to equal its negative. It vanishes in odd characteristic. The symplectic form equation then restricts to each diagonal block, so an element centralising the whole subgroup preserves every symplectic summand. Combining this preservation of the summands with the local centraliser calculation gives the centraliser containment for the direct product.

The remaining radical subgroup factors attached to the polynomial $x - 1$ and their irreducible quotient characters are identified with the construction used in Feng–Malle's Proposition 5.4 [33]. A parameter consists of a subgroup decomposition and an independent staircase parameter for each quotient character. A separate hypothesis classifies these subgroup decompositions up to conjugacy. The character parametrisation concerns the quotients of those subgroups. The resulting maps in the two directions are proved to be inverse.

For the inverse map, assume a factor decomposition of the given principal weight, the subgroup formula and the normalisation of its basic factors. Together with the exclusions, these determine the surviving factors, their symplectic summands and the subgroup identity. No description of the normaliser or enumeration of quotient characters is assumed at this step. The independent signs and the local centraliser calculation above prove centraliser containment for the constructed subgroup. This containment is then used to obtain the normaliser description.

The numerical parameters satisfy the weighted rank condition. Their colours and diagonal action are those in Feng–Malle's local formulae and parameter action [33, Lemma 5.1 and Proposition 5.9]. The comparison maps preserve the specified subgroup, character and conjugator. Thus the field and diagonal actions on the numerical parameters give the actions on principal weight classes.

For principal Brauer characters in rank at least three, manuscript Lemma 2.12 gives the ordinary character selection in $H = \mathbf{H}^{F_0} = \mathrm{CSp}_{2m}(q_0)$, where q_0 is odd. Choose the geometric semisimple element s_0 and Weyl group family $\mathcal{F}$ in [69, 10.1–10.3], with s_0 in a maximal torus of $\mathbf{H}^*$ split over $\mathbb{F}_{q_0}$. Its image in the adjoint group is quasi-isolated [69, 6.19]. That image has square one [10, Proposition 4.11(a)]. Since $Z(\mathbf{H}^*)$ is a torus, choose $z \in Z(\mathbf{H}^*)$ with $(zs_0)^2 = 1$ and set $\tilde{s} = zs_0$. Multiplication by z leaves the centraliser and its Weyl group family unchanged. Since $\tilde{s}^2 = 1$ and q_0 is odd, F_0 fixes $\tilde{s}$ in this split torus and acts trivially on $W_{\tilde{s}}$. The ordinary family used below is $\mathrm{Irr}_{\tilde{s}, \mathcal{F}}(H)$ for this rational pair.

The construction in [69, Proposition 10.2] identifies truncated induction of the special character of $\mathcal{F}$ with the Springer character of the chosen unipotent class $\mathcal{C}$. The family group and the component group $A_{\mathbf{H}}(u) = C_{\mathbf{H}}(u)/C_{\mathbf{H}}(u)^\circ$, for $u \in \mathcal{C}$, are elementary abelian of the same order, as established in the manuscript using [69, 7.1 and Proposition 10.2]. The argument in [69, 7.4] applies to $\tilde{s}^2 = 1$ and gives condition (P4) of [69, Theorem 2.11]. This supplies the geometric hypothesis [41, (4.4)(b)] used in place of isolation.

The proof, in Appendix B of the manuscript, applies the geometric argument of Lemma 2.11 to these Type C families. An auxiliary Frobenius $F' = F_0^{2N}$, with N sufficiently divisible, satisfies [41, (4.7)(a)–(c)] and has field size $q' = q_0^{2N} \equiv 1 \pmod{4}$. The diagonal and support identities, the comparison of Fourier constants and the multiplicity calculation establish [41,

Proposition 3.1 and Corollary 3.2] for the groups required in the argument. The constants needed for the restriction theorem then follow from [41, Theorem 3.8]. The resulting restriction multiplicities are independent of the rational structure. For the original F_0 , the local system trace values give the character table of $A_H(u)$ up to nonzero row factors. Shoji’s theorem [64, Theorem 3.2] expresses the corresponding character sheaf trace functions as nonzero scalar multiples of almost characters in the same ordinary family. Thus ordinary values from that family span all functions on the rational classes in $\mathcal{C}^{F_0}$.

For this original Frobenius, triviality of its action on $W_{\tilde{s}}$ and the abelian family group give the generic denominator formula [41, (6.1)(c)]. Together with the weighted scalar product formula and nonnegative integral multiplicities, it makes every scalar product row have sum one. The spanning and independence arguments show that every column has a nonzero entry. This gives the required Kronecker delta scalar products with the generalised Gelfand–Graev characters.

Proposition 3.3 descends this selection to $G_0 = \mathrm{Sp}_{2m}(q_0)$. The nondegenerate symbols in each type D factor of the selected family allow the argument of [69, 10.3] to give condition (P5) of [68, Theorem 3.2] for the same split pair. The restriction argument in [68, Proposition 5.4] uses this condition, induction of generalised Gelfand–Graev characters [70, proof of Lemma 14.12], compatibility with Alvis–Curtis duality and Frobenius reciprocity. It relates the restriction constituents to the rational classes of G_0 in each H -class. Their unipotent supports are preserved under restriction [45, Example 2.7.18]. Their rational Lusztig series has parameter the image of $\tilde{s}$ [45, Proposition 2.6.16]. This image has square one, so the constituents and their positive Alvis–Curtis duals belong to the principal block [21, Proposition 9.8(iv) and Theorem 21.14]. The vanishing results for wave front sets give the zero entries needed for unitriangularity in [24, Theorem 2.9]. This is the character selection used in the principal block argument of [33, Section 4.1 and Corollary 4.3].

The total numbers and the numbers fixed by the diagonal action follow by combining the Brauer character parametrisation with the counting identities in [33, Corollary 4.3, Proposition 6.1 and proof of Theorem 6.2]. Inner automorphisms and field automorphisms act trivially on the sets being compared. The remaining diagonal action is involutory on those sets. The general fixed point count argument in the library manual therefore constructs the equivariant bijection.

Implementation.

The main files are `ManuscriptIBAW.TypeC.PrincipalConverse`, `ManuscriptIBAW.TypeC.PrincipalProducts`, `ManuscriptIBAW.TypeC.PrincipalParameters`, `ManuscriptIBAW.TypeC.PrincipalNumericalParameters`, `ManuscriptIBAW.TypeC.PrincipalParameterCoordinates` and `ManuscriptIBAW.TypeC.PrincipalParameterActions`. The structures `PrincipalFactorData` and `PrincipalFactorSelection` in `PrincipalParameters` specify the factor decompositions used by the inverse construction. `ManuscriptIBAW.TypeC.PrincipalEquivariance` constructs the correspondence from the specified actions and counts. In `ManuscriptIBAW.TypeC.GGGRSelection`, `delta_selection` constructs distinct characters in the specified ordinary family of CSp_{2m} whose positive Alvis–Curtis duals have the required Kronecker delta scalar products. In the application, this is the family indexed by the centrally normalised split pair $(\tilde{s}, \mathcal{F})$ above. The construction of that pair, condition (P4) and the geometric interpretation of the family are not formalised. The scalar products are normalised by the group order and use inverse character values, with trace identities in the same convention. The trace expansions for the original Frobenius, support and duality identities, nonnegative multiplicities, weighted formula, denominator equality and trivial component action are explicit assumptions. Their geometric proofs, including the auxiliary Frobenius argument, remain outside the formalisation. Lean derives spanning, independence, localisation of scalar products and a nonzero entry in every column. It also proves that the component indices are one and every row has sum one.

`ManuscriptIBAW.TypeC.PrincipalGGGR` selects characters of the symplectic group using the specified restriction identities, Frobenius reciprocity and Clifford action. In `P5Source`, `p5_count` assumes the following numerical consequence of condition (P5) and the class count in [68, Proposition 5.4]: whenever the scalar product on H is nonzero, the number of constituents on restriction to G_0 equals the number of G_0 -conjugacy classes in the corresponding H -conjugacy class. Lean proves the column sums from the restriction identities. Clifford transitivity and the conjugation identities make all row sums equal, and the assumed count forces them to be one. This selects the required characters of G_0 . Geometric support and wave front vanishing remain explicit assumptions. The finite character selection is a separate result. Its interpretation as ordinary characters in the principal block, and the deduction of principal block Brauer character counts from the selected rows, are not formalised. The principal block application instead assumes `FengMalleCorollary43aSource` and `FMPrincipalCountSource` directly, as the fields `fieldFixed` and counts of `PrincipalGGGR.PrincipalSource`. These are the field invariance and counting conclusions used from Feng–Malle. The application does not derive them from `lowerTriangular`.

The application to all blocks separately assumes [33, Corollary 4.6] as `LiteralFengMalleCorollary46Certificate`. It is not a consequence of the principal character selection. The covering and character triple hypotheses also remain explicit.

For the application to all blocks in rank two, `ManuscriptIBAW.TypeC.LiLiSource` takes Li–Li’s theorem as an external result for each block at a nondefining prime over an odd field, in the specified splitting modular system. Its application requires an identification of the acting group with the full automorphism stabiliser of that block. This identification is the hypothesis `Definition35AutomorphismStabilizerAdapter`, and is not a conclusion asserted for an arbitrary group action. The cover is the identity on PSp_4 at two and the map $\mathrm{Sp}_4 \rightarrow \mathrm{PSp}_4$ at odd primes. `ManuscriptIBAW.TypeC.LowRankApplications` combines these blockwise conclusions using a separate compatibility assumption across blocks.

The Type C principal family uses an algebraically closed ordinary coefficient field. The source `ClosedSource` interprets Li–Li’s theorem directly in this field, with the block and root conventions in `ClosedCoefficients`. This interpretation is an external assumption. `ClosedRankTwoPrincipalInputs.seed` lifts the resulting quotient block bijection to Sp_4 . For each Brauer character and its corresponding weight, a separate assumption identifies the central quotients defined by the reference character and the given Brauer character and supplies compatible roots of unity, extensions and intermediate block data. The assumptions `SingleBlockCompatibilitySource` and `ClosedSingleBlockCompatibilitySource` apply to every coefficient prime, family, cover and coefficient choice of the specified types, every block and block witness, and every Brauer character in that block. They are not restricted to rank two and concern the specified universal ℓ' -covers. Each supplied block witness already identifies its acting group with the full automorphism stabiliser of the block. These applications use the instances selected by the central quotient construction. The given bijection and its equivariance are preserved. The central lift retains its inflation and character triple hypotheses.

For the modular character triple and covering arguments, choose the local characters in a common root convention. An adjustment within diagonal orbits gives a correspondence satisfying the required triple relation for every character and its weight. The extension and covering steps use [33, Proposition 3.4, Remark 3.5 and Corollary 4.6] with those compatible choices.

3.2 The unipotent correspondence over even fields

For even q , Li [52, Section 6] obtains the descriptions of blocks and weights by adapting arguments of An and Fong–Srinivasan. Li proves the inductive condition under a unitriangularity assumption on the relevant decomposition matrix [52, Theorem 3]. In rank at least four, the manuscript gives an independent proof using generic weights and Jordan reduction, without that assumption.

Let $H = \mathrm{Sp}_{2r}(2^a)$ with $a > 0$, let ℓ be odd, and let e be the multiplicative order of 2^a modulo ℓ . For a unipotent block b , let X_b be its unipotent ordinary characters and let $\mathcal{W}(b)$ be the set of conjugacy classes of generic weights belonging to b .

The unipotent block parametrisation and the relative Weyl group argument must use the same pair. Cabanes–Enguehard’s Theorem 4.4 identifies the relevant series [20]. We use the relative Weyl group parametrisation recalled in Cabanes–Späth’s Theorem 2.8 [22]. Kessar–Malle’s Definition 2.1, Remark 2.2(2) and Theorem A give the comparison with the required e -Jordan-cuspidal pair in this setting [49]. The uniqueness and block membership statements identify this pair and establish that it is e -Jordan-generalised-cuspidal.

Theorem 7.5 of Feng–Malle–Zhang then gives the relative Weyl group correspondence [34]. Their Lemma 3.21 concerns one pair. Passing to all generic weights also uses Definitions 3.17–3.18, the conjugacy classes of pairs and the character correspondence on their inertia quotients. The passage to all weights is conditional on the required extendibility for every pair, and the admissible pairs are explicitly e -split. The resulting character correspondence also gives the cardinality equality.

The field invariance argument uses a rational Levi parametrisation and the same Lang conjugator throughout. Let $\mathbf{M}$ be the selected rational Levi subgroup and let F' define H . The field automorphism of a standard Levi is transported to an automorphism τ fixing the selected pair. The action of τ on $N_H(\mathbf{M})/\mathbf{M}^{F'}$ is trivial. Extend the unipotent character to the required stabiliser using [67, Theorem 3.3]. This application uses the embedding of $\mathbf{H}$ into the simple adjoint group PGL_{2r+1} that sends A to the projective image of $\mathrm{diag}(A, 1)$. The extension group is the simultaneous stabiliser of the subgroup and its unipotent character, including the prescribed field automorphisms. The geometric hypotheses identify this stabiliser and the inclusion of the finite Levi subgroup. Gallagher’s theorem and induction then give invariance of the generic weight class. The ambient Frobenius powers are required to induce the specified automorphisms on the finite fixed point groups. No corresponding finite order relation on the entire algebraic group is assumed.

The integral basic set in [39, Theorem A] identifies the free abelian groups on X_b and $\mathrm{IBr}(b)$ through the specified decomposition homomorphism. Field automorphisms fix X_b pointwise, so the elementary basis argument proves that they also fix $\mathrm{IBr}(b)$ pointwise. Equality of the ranks gives an equivariant bijection between these sets.

The complex ordinary characters and the ordinary labels in the basic set are related through a common cyclotomic field. The two fixed embeddings identify the values of the same character and therefore determine the automorphism action on its label over the ordinary coefficient field. The comparison of labels thus commutes with the given automorphisms.

Under Condition 6.1, Theorem 6.2 of [34] gives an equivariant correspondence between generic weights and Alperin weights, preserving blocks. Here ℓ is good and does not divide $2 \cdot 2^a |Z(H)|$. The regular embedding and automorphism group are those specified in that condition. Manuscript Lemma 2.10 applies the specialised Brough–Späth criterion, in the form stated in [31, Theorem 3.18], to show that the block is BAW-good. The passage from BAW-goodness to the modular character triple condition uses the stated identifications of the local character reductions and compatible roots.

Implementation.

`ManuscriptIBAW.TypeC.PairLabelComparison` identifies the initial pair.

`ManuscriptIBAW.Characters.CyclotomicOrdinaryLabels` compares character values over the two coefficient fields. The direct basic set and unipotent arguments are in

`ManuscriptIBAW.TypeC.EvenBasicSet` and `ManuscriptIBAW.TypeC.EvenUnipotent`. The local semisimple label comparison in manuscript Lemma 3.6 is an explicit external assumption, as are the rational Levi parametrisation and the extension, Clifford and Gallagher formulas. Given these

statements, the formal proof derives field invariance of the generic weight classes. It does not formalise the Deligne–Lusztig theory underlying the label comparison. The modules beginning `ManuscriptIBAW.TypeC.EvenApplication` apply the criterion to the specified groups and covering maps under the stated source hypotheses.

3.3 Jordan reduction for each semisimple parameter

Jordan reduction uses every pair $(\mathbf{K}, F')$ required by Feng–Li–Zhang’s Hypothesis 5.5 [31], including the pair to which the reduction is being applied. The corresponding group $\mathbf{K}^{F'}$ is included even if it has a nontrivial central ℓ -subgroup or is not itself a universal ℓ' -cover.

The automorphism group in the stabiliser and extension hypothesis is chosen separately for each semisimple parameter s , in accordance with the construction in Section 5 and Remark 5.4 of the same paper. Restrict the stabiliser factorisation and a character extension to the chosen subgroup, retaining the representation affording the same character. Irreducibility after restriction follows because the restricted group still contains the base group on which that representation is irreducible.

The geometric choices identify field automorphisms representing the required outer automorphisms stabilising the series of s . Compatible Frobenius representatives and a Levi subgroup in duality with the minimal dual Levi are chosen by the construction using inner conjugacy in [60, Corollary 4.2, Lemma 4.5 and Proposition 4.6], preserving the same series idempotent and the required outer stabiliser. These choices do not assert stability of an arbitrary preselected Levi. The stabiliser and extension hypotheses for this chosen group are established before applying Theorem 5.7.

The geometric series use the original blocks. If t_b is the semisimple ℓ' -parameter assigned to b , the series idempotent at s is identified with the sum of the primitive idempotents whose parameters are conjugate to s in the finite dual group. Orthogonality proves that a block belongs to the series exactly when this conjugacy condition holds. The specified inclusion of finite rational points in the algebraic dual sends this same t_b to the parameter used in the strict quasi-isolation argument. Thus the block of the lower group and its geometric series refer to the same parameter before the reduction theorem is applied. The interpretation of the supplied series, dual group and semisimple elements remains an external hypothesis.

For odd q and $\ell = 2$, the applicable quasi-isolation statement forces the semisimple odd-order label to be the identity, and the lower symplectic block is principal [10, Proposition 4.11(a)]. For characteristic two and odd ℓ , the quasi-isolated label is also the identity, so the block is unipotent [10, Example 4.8]. These applications require the stated semisimplicity, dual group type and characteristic conditions.

For a rank two relative factor over an odd field, the manuscript uses the principal Sp_4 block bijection obtained from Li–Li’s theorem by the central lift in Proposition 3.3. For higher rank factors it uses the principal construction above.

The lower type A cases use the special linear or unitary fixed point groups, including their central ℓ -subgroups when present. The invoked result retains the type A hypotheses [32] and those for the exceptional small groups [29]. For split low ranks over even fields of order at least four, Schaeffer Fry’s Theorem 5.5 applies. For $\mathrm{Sp}_6(2)$ it applies at every odd prime on the specified double cover [63]. The Suzuki cases use [65, Corollary 6.3], together with Proposition 4.6 of the same paper for the required descent to the fixed point group. The specified covers are retained, including the exceptional cover for ${}^2B_2(8)$.

Implementation.

The general restriction and geometric constructions are in `ManuscriptIBAW.Jordan`, including the block and label identities in `ManuscriptIBAW.Jordan.SeriesBinding`. The specialised applications are `ManuscriptIBAW.Jordan.TypeCOddLocalized` and `ManuscriptIBAW.Jordan.TypeCEvenLocalized`. `ManuscriptIBAW.TypeC.PrincipalApplicationComplete` and `ManuscriptIBAW.TypeC.LowRankApplications` keep the corresponding complete family and the specified groups in small rank. Rank two principal relative factors use the specified Li–Li application and central lift. Higher rank factors use the principal construction with its stated character theoretic assumptions.

3.4 Odd nondefining primes over odd fields

Let $\tilde{G} = \mathrm{CSp}_{2n}(F)$, let E be the cyclic group of field automorphisms, and let C be the group of ordinary linear characters of $\tilde{G}/G$ of order prime to ℓ . The multiplier identifies $\tilde{G}/G$ with $F^\times$ and sends the centre onto its squares. Thus $|\tilde{G} : GZ(\tilde{G})| = 2$ when q is odd.

For a block stabiliser J in $C \rtimes E$, the kernel D of the field projection embeds in its linear character stabiliser. The general central character argument gives $|D| \leq 2$, and every subgroup of J is 2-hypoelementary. The multiplier and central character calculations therefore give both the quotient index and the block stabiliser bound.

The required integral basic set is given by [43, Theorem 5.1]. Its decomposition homomorphism, root convention and actions remain specified. Conlon’s theorem [25], in the formulation of [12, Theorem 3.5.5], gives the equivariant comparison needed here. The prime used for the permutation lattice is 2, while ℓ is odd.

For the symplectic group, the ordinary basic set is given by Theorem A [39]. Here $\mathcal{E}_{\ell'}(G)$ is defined from the rational Lusztig partition on complex characters by requiring a semisimple dual parameter of order prime to ℓ . The dual groups are the split special orthogonal group for G and the special Clifford group for $\tilde{G}$. Fixed embeddings of one cyclotomic field compare the same character values over $\mathbb{C}$ and the ordinary coefficient field. The sets used in the two basic set arguments are assumed to be these unions of rational series. Their identification with the Lusztig partition in the cited results remains an explicit hypothesis.

Cabanes–Enguehard Proposition 15.6(i) gives the restriction comparison for each fixed rational semisimple lift of a parameter from the special orthogonal dual to the special Clifford dual [21]. Membership in the lower series is equivalent to occurrence in the restriction of an upper character in that fixed upper series. A separate structural assumption supplies the semisimple lift under the specified projection from the special Clifford group to the special orthogonal group. Reindexing the ordinary restriction scalar product proves that conjugation by $\tilde{G}$ preserves occurrence as a restriction constituent. The same upper lift and upper character then prove diagonal stability of the lower series. The lower parameter is unchanged, so its order remains prime to ℓ . This argument does not require an upper lift of order prime to ℓ . The source interpretation uses the specified regular embedding, equality of derived groups, connected centre of the conformal algebraic group and connected central dual kernel.

For the field action, let σ be the specified field automorphism of G , and let σ^* denote the specified coordinate field automorphism of the split orthogonal dual. Taylor’s Proposition 7.2 transports χ by pullback $\chi \mapsto \chi \circ \sigma^{-1}$ and the label s by $(\sigma^*)^{-1}(s)$ [71]. The algebraic dual isogeny identification and the common torus and root conventions remain part of this ordinary source statement. The inverse dual automorphism preserves order, so semisimplicity

and order prime to ℓ are preserved. The common cyclotomic character identities give the same conclusion over the ordinary coefficient field.

These calculations prove stability of the union under the full conformal and field action. Field automorphisms may permute its individual rational series. On this union, the remaining basic set assumptions give the integral decomposition homomorphism and its compatibility with blocks. Apply the block basic set argument over representatives of block orbits and transport the resulting bijections over their orbits. This constructs one bijection preserving blocks

$$\beta : \text{IBr}(G) \longrightarrow \mathcal{E}_{\ell'}(G)$$

commuting with the conformal and field actions. Surjectivity of their action onto $\text{Aut}(G)$ gives equivariance under every automorphism.

The ordinary character and weight formulae in [52, proofs of Lemmas 5.1 and 5.3] describe the same changes of labels under tensoring and field automorphisms. The unitriangularity assumption in those lemmas is used to transfer ordinary labels to Brauer characters. The comparison of ordinary character and weight labels used here does not require that assumption. The ordinary stabiliser factorisation is supplied by [23, Theorem 3.1].

Apply that ordinary theorem to $\beta(\varphi)$. Equivariance and injectivity of β transfer the factorisation of conformal and field stabilisers to every $\varphi \in \text{IBr}(G)$. The common character identities transport the ordinary theorem from $\mathbb{C}$ to the coefficient field. The restriction expansion can then select a constituent with the required stabiliser.

Li proves the weight stabiliser factorisation for the standard radical subgroups described in [52, Section 2.C], in [52, Lemma 5.5(1)]. Conjugating the chosen representative gives the factorisation for every weight. The manuscript also uses [52, Lemma 5.5(2)] for the ordinary character extension in condition (iv) of [19, Theorem 4.5].

The character and weight stabiliser factorisations, cyclic extensions and subgroup index calculations give the hypotheses of the criterion in [19, Theorem 4.5] on the specified symplectic cover. The passage to the complete family condition also uses [50, Lemma 3.3] and the normalisation in [65, Definition 4.1], with the specified roots and block operations.

Implementation.

The conformal correspondence is `ManuscriptIBAW.TypeC.OddConformalCorrespondence`. `ManuscriptIBAW.TypeC.OddRationalSeries` defines the two rational unions. `ManuscriptIBAW.TypeC.OddRationalAutomorphisms` proves stability of the symplectic union under diagonal and field automorphisms from the stated ordinary character formulae. `ManuscriptIBAW.TypeC.OddLiteralBasicSet` combines this stability with the integral decomposition hypotheses. The assumptions are given by `OddPrimeApplication.SourceInputs.basicSetData`, from which `basicSet` is constructed. `ManuscriptIBAW.TypeC.OddGlobalFactorization` constructs the global bijection and transfers the stabilisers. `ManuscriptIBAW.TypeC.OddPrimeApplication` derives the index consequences and applies the criterion. Its field `SourceInputs.li` has type `TypeCCurrentOddPrimeLiSource.Li55SplittingCertificate`. This external assumption requires the standard radical subgroup family and its weight stabiliser factorisation over every coefficient field of characteristic zero containing the required roots of unity for the full semidirect product. The ordinary character extensions use separate cyclic extension assumptions on the two inertia quotients. The Brauer character extension uses a separate assumption. `ManuscriptIBAW.TypeC.HigherOddApplication` specifies the cover and simple quotient before invoking that result.

Chapter 4

The inductive condition for Type B

Let $\mathbb{F}_q$ be a finite field of order $q = p^f$. Except in the final discussion of small ranks and even characteristic, assume that q is odd, $n \geq 3$ and $S = \Omega_{2n+1}(q)$ is the specified nonabelian simple matrix group. The spin and special Clifford groups are formed from the same quadratic space. Their homomorphisms to the orthogonal groups are the specified conjugation maps.

The proof separates the defining prime, odd nondefining primes and the prime two. For $\ell = 2$, the main steps are the principal block argument using generalised Gelfand–Graev characters and the passage from Levi subgroups to all blocks by Jordan reduction. The group $\Omega_7(3)$ requires a separate covering group. The final case distinction includes small ranks and even characteristic. The structural identifications and the cited geometric and character theoretic results remain assumptions where indicated. The geometric argument in manuscript Lemma 2.11 is also unformalised. Chapter 1 explains the scope of the formal verification.

4.1 The defining prime and its covering group

At $\ell = p$, the required cover is $G = \text{Spin}_{2n+1}(q)$ with its projection $G \rightarrow S$. The Clifford construction proves surjectivity and identifies the kernel with the centre under the specified structural assumptions. The centre has order two, hence order prime to the odd prime p . Perfectness, simplicity of S and maximality among perfect central extensions with kernel of order prime to p are the remaining covering assumptions.

Späth’s Theorem C is then applied on this cover [65, Theorem C and its proof]. The source formulation uses the specified blocks, root correspondence, local characters and block operations, and concludes the prescribed complete family condition. The family and covering map are defined before the theorem is applied. Its conclusion includes the complete character triple and extension requirements.

This also treats $S = \Omega_7(3)$ at $p = 3$. The relevant cover is $\text{Spin}_7(3) = 2.\Omega_7(3)$, although the full universal central cover has kernel of order six [56, Tables 24.2–24.3 and Remark 24.19]. Only the universal p' -property is required. The ordinary coefficient field K has characteristic zero. This application does not require it to be the fraction field of a discrete valuation ring. The algebraically closed field k of characteristic p and the root and coefficient conditions of the published source, remain specified separately.

Implementation.

`ManuscriptIBAW.TypeB.Defining.SourceInputs` states the assumptions on the same spin group and projection. In that namespace, `complete` combines the complete family supplied by the published theorem with `D.facts.actualCover`, the covering property of the specified spin projection.

4.2 Odd nondefining primes

Let $\ell \neq p$ be odd, let $G = \text{Spin}_{2n+1}(q)$, and let $\tilde{G}$ be the special Clifford group. The Clifford norm sends an element g to the scalar represented by $g \text{ rev}(g)$, where rev denotes reversal in the Clifford algebra. The group G is its kernel. Surjectivity is proved by taking a product of two vectors with prescribed nonzero norms. We assume that the centre consists of scalar units. Their norms are squares, so the general theorem on multiplier quotients gives

$$\tilde{G}/GZ(\tilde{G}) \cong \mathbb{F}_q^\times / (\mathbb{F}_q^\times)^2, \quad |\tilde{G} : GZ(\tilde{G})| = 2.$$

The larger quotient $\tilde{G}/G$ has order $q - 1$.

Let C be the group of ordinary linear characters of $\tilde{G}/G$ of order prime to ℓ , let E be the finite cyclic group of field automorphisms, and set $A = C \rtimes E$. Tensoring acts on a primitive idempotent by scaling the group algebra basis at g by the corresponding modular character value. With this left action, the module on which the idempotent acts as the identity is tensored by the reciprocal character. Cancellation in the group algebra proves this support equation on the same idempotent and module. The ordinary lifts and modular characters are compared using the common roots and their residue compatibility.

For a block b with stabiliser $J = A_b$, the kernel of the field projection embeds in this tensor stabiliser. The central character bound and the preceding index calculation make its order at most two. It is normal, and the quotient is cyclic. Every subgroup of J is therefore 2-hypoelementary. The identification of the centre with scalar units is a structural assumption. The index, compatibility with module support and stabiliser bound follow from it by the preceding arguments.

For $M \in \{G, \tilde{G}\}$, the integral basic set theorem in the form stated in Feng–Li–Zhang’s Theorem 2.3 gives an integral basic set on each rational ℓ' -series [28]. The odd nondefining prime is good, and the component groups of the algebraic centres have orders two and one, respectively. The decomposition into unions of blocks and the block diagonal form of the decomposition map identify its restriction to a block b of M as

$$d_b : \mathbb{Z}[\mathcal{X}_b] \xrightarrow{\sim} \mathbb{Z}[\text{IBr}(b)], \quad \mathcal{X}_b = \text{Irr}(b) \cap \mathcal{E}(M, \ell').$$

The action preserves these ordinary characters and commutes with the specified reduction. The interpretations of the rational series and ordinary character values, including stability under tensoring [45, Theorem 4.7.1(3)], remain supplied on the specified dual groups and common character parameters. This argument requires an integral basic set, without an additional unitriangularity hypothesis.

On G , the effective outer action is the diagonal group of order two times the finite cyclic group of field automorphisms. Its block stabilisers are also 2-hypoelementary. Conlon’s permutation lattice theorem [25], in Bouc’s formulation [12, Theorem 3.5.5], gives equality of the subgroup marks. Burnside’s injectivity theorem for marks [12, Theorem 2.3.2] then gives the equivariant bijections in manuscript Lemma 4.2. These results apply to the decomposition lattice. The prime used by Conlon is two, while ℓ is odd.

For $\tilde{G}$, compose the constructed block bijection with the separately published correspondence from ordinary characters to weights [28, Proposition 7.2]. Choose these maps on representatives of block orbits and transport them over the orbits. This constructs the global A -equivariant correspondence between Brauer characters and weights required by the criterion. It preserves blocks.

For G , the construction over block orbits first gives one global equivariant bijection between all irreducible Brauer characters and the selected ordinary series. Apply the ordinary character

stabiliser theorem of Cabanes and Späth, in the form recalled in [28, Theorem 3.10], to its image. Equivariance and injectivity transfer the stabiliser factorisation back to every Brauer character. The ordinary theorem has no Brauer or unitriangularity premise. Only after this transfer does the restriction expansion select a constituent of an upper Brauer character. The normaliser statement for weight representatives is supplied by Proposition 7.5 of the same paper.

The remaining criterion clauses use the same cyclic quotient, inertia groups and local characters. Cyclic extension principles, with their root hypotheses, supply the Brauer and local ordinary extensions. Both inertia groups contain $GZ(\tilde{G})$. The index two and oddness of ℓ then give the required equality involving the inertia group and the preimage of a Hall subgroup of $\tilde{G}/G$. This equality is proved for the specified characters and weight classes.

The manuscript applies the criterion in Feng–Li–Zhang’s Theorem 2.1 [28]. The formal argument combines Brough–Späth’s criterion [19, Theorem 4.5], Koshitani–Späth’s passage from block conditions to compatible correspondences [50, Lemma 3.3] and Späth’s full condition [65, Definition 4.1]. The roots of the common modular system and coefficient normalisation remain prescribed. This combined statement is not asserted to be the verbatim conclusion of Theorem 2.1. The family, local reductions and generic spin cover are fixed before the full inductive condition is deduced. The structural assumptions include the universal ℓ' -property. They do not separately formalise the full universal central extension assertion about the spin group.

For $(n, q) = (3, 3)$, use the sixfold cover $6.\Omega_7(3)$. The exact order and prime conditions give $\ell \in \{5, 7, 13\}$, each occurring to exponent one. Every ℓ -subgroup is therefore cyclic. The complete result for cyclic defect is applied on this cover [50, Theorem 1.1]. The identification of the simple group with the matrix group $\Omega_7(3)$ transfers the complete condition to that group while retaining the same cover and characters.

Implementation.

`ManuscriptIBAW.TypeB.OddTensorBound` proves the index formula determined by the norm and centre, and the tensor stabiliser statements. `ManuscriptIBAW.TypeB.OddConlon` constructs the upper block bijections. `ManuscriptIBAW.TypeB.OddApplication` constructs the correspondence and criterion hypotheses on its fixed family. `ManuscriptIBAW.TypeB.OddUniform` selects the generic or exceptional cover, and `OddInputs.Inputs.complete` provides the certificate on the matrix orthogonal group. These applications use one modular system and the required finite ordinary roots. Its fraction field is not required to be algebraically closed.

4.3 Fixed representatives in products

Manuscript Lemma 4.4 is a finite product argument. Let $N = \prod_{i \in I} H_i$, let $\Delta = \prod_{i \in I} \Delta_i$ act componentwise, and let a cyclic group E permute the factors and their actions. For a product character, consider its Δ -orbit $\mathcal{O}$ and choose a generator τ of its setwise stabiliser $E_{\mathcal{O}}$.

On each cycle C of the permutation induced by τ , choose one component representative whose stabiliser factors with respect to Δ_i and the iterate $\tau^{|C|}$ acting on the chosen factor. Stability of its orbit allows that iterate to be corrected by a diagonal element. The factorisation then implies that the chosen representative is fixed by the iterate itself. Propagate it around the cycle by powers of τ . Fixedness under $\tau^{|C|}$ makes these choices consistent when the cycle closes. Each coordinate remains in its original component orbit, and the resulting product character θ is fixed by τ , hence by $E_{\mathcal{O}}$. Consequently its stabiliser in $\Delta \rtimes E_{\mathcal{O}}$ is $\Delta_{\theta} \rtimes E_{\mathcal{O}}$.

The proof needs positive finite cycle lengths and the stated generator property. It does not need E to be finite. In a character application, the product parametrisation, orbit identification

and equation for the iterate on the chosen coordinate must concern the same characters and actions. That equation also identifies the iterate with the stated field automorphism.

Implementation.

`ManuscriptIBAW.TypeB.Levi` exports `component_return_full` and `component_return_full_of_product_orbit_stable` from `ModularRep.ComponentReturnFull`. The latter obtains the orbit transport needed in the construction from stability of the product orbit.

4.4 Rational Levi factors and their character orbits

For manuscript Lemma 4.5, fix the regular embedding $\mathbf{G} \hookrightarrow \tilde{\mathbf{G}}$ and Frobenius F of the statement, an F -stable Levi subgroup $\mathbf{L}$ of $\mathbf{G}$, and $\mathbf{H} = [\mathbf{L}, \mathbf{L}]$. Write $\mathbf{H} = \prod_{j \in I} \mathbf{H}_j$ using its specified simply connected components. If C_i is an F -orbit of components, choose $x_i \in C_i$, let $n_i = |C_i|$, and set $H_i = \mathbf{H}_{x_i}^{F^{n_i}}$. The coordinate equations for F construct an isomorphism

$$\mathbf{H}^F \cong \prod_i H_i.$$

An element fixed by F^{n_i} determines all coordinates around its component orbit, and these assignments give the two inverse maps. The geometric component presentation is an assumption. The rational product map is deduced from it. For the component decomposition, see [56, Proposition 12.14]. For its relation to rational points, see [45, Lemma 1.5.15 and Corollary 1.5.16].

Let $\tilde{\mathbf{L}} = \mathbf{L}Z(\tilde{\mathbf{G}})$ and let Δ_i be the image of its rational conjugation action on H_i . The proof also constructs surjectivity of

$$\tilde{\mathbf{L}}^F \longrightarrow \prod_i \Delta_i.$$

Choose an element of $\tilde{\mathbf{L}}^F$ inducing the desired action on one factor and write it as hz , with $h \in \mathbf{H}$ and z central. Project h onto the corresponding full Frobenius orbit of components. This projection commutes with F . Because hz is rational, the projected element is fixed by F up to a central factor. The Lang–Steinberg theorem on the connected centre supplies a central correction making the element rational, with the prescribed action on the chosen rational factor and trivial action on all other factors. Taking the product of these elements over the factors proves surjectivity.

The geometric decomposition used here is $\tilde{\mathbf{L}} = \mathbf{H}Z(\tilde{\mathbf{L}})$. Its rational analogue is not assumed. The centre must be connected and stable under the specified Steinberg map before the central Lang equation is justified by Geck–Malle Theorem 1.4.8 [45]. The proof gives the product of the conjugation images. Identifying them with the full groups of inner and diagonal automorphisms also requires identifications of the adjoint groups and their surjective rational point maps. The required comparison for a regular embedding is described in Geck–Malle Remark 1.7.6(a). These assumptions concern the adjoint groups and maps of the specified algebraic groups.

We use the standard parametrisation of irreducible Brauer characters of a finite direct product [58, Theorem 8.21]. The parametrisation, product value formula and compatible root lifts are assumptions of this argument. Evaluating the product formula shows that the parametrisation commutes with automorphisms. Surjectivity of the preceding action map then shows that a $\tilde{\mathbf{L}}^F$ -orbit of Brauer characters of $\mathbf{H}^F$ is exactly the Cartesian product of the component orbits.

Normality of $\mathbf{H}^F \triangleleft \mathbf{L}^F \triangleleft \tilde{\mathbf{L}}^F$ and the abelian quotient follow from the same geometric subgroups. An injective field endomorphism commuting with F restricts to an automorphism of the finite

rational group. Its preservation of the Levi and centre therefore gives setwise preservation of this normal chain. No finite order relation is imposed on its action on all algebraic points.

Implementation.

`ManuscriptIBAW.TypeB.Levi` exports `rationalProduct`, `imageProduct_surjective` and `cartesian_character_orbit` from `ModularRep.PaperProofs.TypeBRegularLeviOrbitAssembly`. The hypotheses supply geometric component coordinates, solutions of the central Lang equations and character products compatible with the chosen roots.

4.5 Clifford transfer of a stabiliser factorisation

Manuscript Lemma 4.6 allows any prime ℓ . Let $N \leq H \leq G$ be finite groups, with $N \triangleleft G$ and G/N abelian. Let a finite group E act on G , preserving N and H . For $\psi \in \text{IBr}(H)$, choose an irreducible constituent θ of ψ_N . Suppose that θ extends to its full inertia group G_θ and that

$$(G \rtimes E)_\theta = G_\theta \rtimes E_\theta.$$

Then

$$(G \rtimes E)_\psi = G_\psi \rtimes E_\psi.$$

The key step is that G_θ fixes ψ . Clifford correspondence expresses ψ as the induction of a character over θ . Gallagher's description writes that character as the restricted extension of θ times a character of the quotient by N . The quotient is abelian, so this latter irreducible Brauer character is linear. Conjugation by G_θ fixes this linear character and the restricted extension, since the extension is a class function on G_θ . Induction commutes with conjugation, so G_θ fixes ψ . One then adjusts the chosen constituent within its H -orbit and applies the assumed factorisation at θ to obtain the factorisation at ψ .

We use the formulations of Clifford theory and Gallagher's theorem in [58, Corollary 8.7, Theorem 8.9 and Corollary 8.20]. Their supplied forms retain the original groups, characters and root comparisons. They do not assume that G_θ fixes ψ . Neither cyclicity of E nor an assumption that the quotient acting on the characters has exponent two is used.

In the Levi application for $\ell = 2$, multiplicity freeness first supplies the extension of θ to its full inertia group. Choose an ambient irreducible Brauer character over θ and use homogeneous Clifford restriction. Multiplicity freeness makes its multiplicity equal to one. Geck–Malle Theorem 1.7.15 allows an algebraically closed coefficient field of arbitrary characteristic [45]. Its application still requires the same regular embedding and connected centre. In the representative construction, one ambient element conjugates both ψ and its selected constituent. The constituent relation and the field stabiliser of the character orbit are preserved under this conjugation.

Implementation.

`ManuscriptIBAW.TypeB.Levi` exports the theorem `lemma46` for an arbitrary prime and the conclusion `same_y_representative`, which uses one conjugating element for the character and its constituent. The characteristic two proof uses the abelian quotient and proves the needed inertia fixedness. The classification of the rational Levi factors into their Type A, rank two and higher spin cases remains a separate structural source assumption in the later application.

4.6 Field invariance and the principal series

For the principal 2-block, set $G = \text{Spin}_{2n+1}(q)$ and let $\tilde{G}$ be its special Clifford overgroup. Let F_0 be the split Frobenius over $\mathbb{F}_p$ and set $F = F_0^f$. A geometric unipotent class is stable

under F_0 because the Jordan block sizes in the orthogonal representation are preserved. The argument on component groups in Taylor's Section 2 supplies a representative fixed by F_0 on whose component group F_0 acts trivially [68]. This includes the nonabelian component groups. Abelianness is not an assumption of this step.

The rational classes in that geometric class are therefore parametrised by the ordinary conjugacy classes of the component group [56, Theorem 21.11]. The supplied comparison commutes with F_0 , so every rational class is fixed. Taylor's equivariance formula then proves field invariance of the same generalised Gelfand–Graev characters Γ_u [71, Proposition 11.10]. The source assumption includes triviality of the action on the component group at the chosen representative. This follows from the component calculation together with a change of rational representative. Existence of a well-chosen representative alone would not supply that identity.

The principal series identification is

$$\text{Irr}(B_0(G)) = \mathcal{E}_2(G, 1),$$

by Cabanes–Enguehard Theorem 21.14 [21]. The adjoint dual group has type C. Bonnafé's Proposition 5.3(a) makes every quasi-isolated semisimple label have order dividing four [10]. Such a label gives an ordinary character in the principal block. A strictly quasi-isolated block has a label of odd order, so that label is the identity. This proves uniqueness of the principal strictly quasi-isolated block. The same result applies to the relative factors of rank two used below.

Implementation.

`ManuscriptIBAW.TypeB.PrincipalField` proves that rational classes and generalised Gelfand–Graev characters are fixed, using the component Frobenius and the parametrisation by ordinary conjugacy classes. The identity action is an inner action with inner element one. In `ManuscriptIBAW.TypeB.PrincipalSeries`, `ContextData` supplies the geometric labels, ordinary characters, root, block selector, primitive idempotent and independent source assumptions. `Binding.certificate` proves the principal series statement from the Cabanes–Enguehard and Bonnafé results together with the clauses on character duality. Only then does `Binding.context` construct the context for the generalised Gelfand–Graev characters on those same objects, using the principal series statement just proved.

4.7 The generalised Gelfand–Graev basis

Let $\mathcal{U}$ index the rational unipotent classes of G , and let P_u be the component of Γ_u in the principal block. For a finite reductive group H and $\chi \in \text{Irr}(H)$, write χ^* for the unique irreducible character equal to $D_H(\chi)$ up to sign, where D_H denotes Alvis–Curtis duality. Manuscript Proposition 4.9 constructs characters $\eta_u \in \text{Irr}(B_0(G))$ such that

$$(\langle \eta_u, P_v \rangle_G)_{u,v \in \mathcal{U}}$$

is block lower triangular when representatives in the same geometric class are consecutive and the geometric classes are ordered by nondecreasing dimension, with permutation matrices on its diagonal.

Here the upper group is the special Clifford group $\tilde{G}$, and the lower group is the spin group G . Fix a geometric unipotent class. Write A for the component group of a representative in G , and Z_A for the image of the spin centre in A . The upper component group is $\bar{A} = A/Z_A$. Taylor's component group description [68, Section 2] gives three possibilities. If $Z_A = 1$, the map from lower to upper rational classes is injective. If A is abelian and $Z_A \neq 1$, then $A = Z_A \cong C_2$ and $\bar{A} = 1$. Otherwise A is extraspecial, with centre $Z_A = \langle \vartheta \rangle$ of order two. In the last case $\bar{A}$

is elementary abelian. Its identity corresponds to two lower rational classes, and each other element to one. The component groups, their Frobenius actions and these rational class maps are structural inputs to the formal argument.

We first obtain characters in an upper family attached to an isolated pair. Starting with the geometric isolated pair supplied by Geck–Hézar’s Proposition 2.4 [42], choose its semisimple parameter in an F -stable maximal torus contained in an F -stable Borel subgroup of the dual group. The replacement removes the central factor and chooses the corresponding symplectic element of order dividing two in this torus. It preserves the geometric unipotent class and the attached finite family group, but may change the rational Lusztig series. It is not an assertion of rational conjugacy with the original parameter. In the central case the new parameter has order dividing two.

Let $(s', \mathcal{F}')$ be this pair, where $\mathcal{F}'$ is a Lusztig family [45, Definition 4.1.17], and write $\tilde{\mathcal{F}}'$ for its family of ordinary characters. The Weyl group is taken relative to the chosen torus. Its Frobenius action is trivial under the stated rational structure. This does not require the central torus itself to be split. The component group $\bar{A}$ and the finite group $\mathcal{G}_{\mathcal{F}'}$ are elementary abelian of the same order.

Manuscript Lemma 2.11, stated among the preliminaries and proved in Appendix B, selects characters in $\tilde{\mathcal{F}}'$ whose positive Alvis–Curtis duals have delta scalar products against the generalised Gelfand–Graev characters of the upper rational classes in the chosen geometric unipotent class. Its isolated Type B hypotheses are the ones used here. Its geometric argument applies Geck’s theorem on restrictions of character sheaves [41, Theorem 4.5] using an auxiliary split Frobenius F' chosen as in [41, (4.7)]. The elementary abelian groups of equal order give condition (4.4)(a). Isolation and the equality of these orders give condition (4.4)(b) by Hézar’s criterion [42, Proposition 2.3].

The manuscript verifies Geck’s identities for generalised Gelfand–Graev characters for the required groups and unipotent classes, including those in cuspidal Levi subgroups and adjoint quotients. For a block attached to a maximal torus, $\zeta = \delta = 1$ gives condition (2.4)(b). The comparison of constants in (3.3)(b) follows from Taylor’s Fourier transform argument [70, Lemma 11.3 and Proposition 11.5] and Letellier’s induction identity [51, (6.2.11)], using compatible split cuspidal data with $w = 1$. This comparison also occurs in the proof of [54, Proposition 7.2]. The remaining integrality argument uses the character existence theorem of Hézar and Lusztig [68, Theorem 1.1], the scalar product formula of Lusztig and Geck–Malle in Taylor’s form [70, Proposition 15.4], and the elementary case of a torus. These give [41, Proposition 3.1 and Corollary 3.2] for the required groups. With $q' \equiv 1 \pmod{4}$, [41, Theorem 3.8] determines the constants needed in the restriction theorem.

Shoji’s theorem [64, Theorem 3.2] is used for F' in Geck’s proof and for the given F after the geometric restriction statement has been established. The geometric argument is proved in the manuscript and is not formalised in Lean.

The local systems are irreducible equivariant $\overline{\mathbb{Q}}_{\ell_0}$ -local systems, with $\ell_0 \neq p$. On a unipotent class they are parametrised by irreducible characters of its component group. A chosen Frobenius isomorphism gives their trace functions. The characteristic functions of character sheaves are likewise Frobenius trace functions. For the given F , Shoji’s theorem [64, Theorem 3.2] identifies the relevant character sheaf trace functions with almost characters up to nonzero scalars. These almost characters lie in the span of the same ordinary family. The finite trace expansions therefore imply that the ordinary character values on the chosen rational classes span all functions on those classes.

The proof uses Lusztig’s modified generalised Gelfand–Graev class functions, written as in Taylor’s equation (11.15) [70]. Taylor’s expansion formulas [70, Theorem 11.13 and Lemma 11.16]

give the scalar products in [40, Lemma 3.5 and Corollary 3.6]. Together with Lusztig's duality identities [54, Proposition 8.5 and Corollary 8.6], applicable by Taylor [70, Corollary 13.6], these give diagonal pairings with nonzero diagonal entries. The diagonal pairings prove that the restrictions to the chosen class of the duals of the modified functions are linearly independent. Their expansions express these restrictions as linear combinations of the restrictions of the Alvis–Curtis duals of the generalised Gelfand–Graev characters. Since the two families have the same size, these restricted duals are linearly independent.

The support of the Alvis–Curtis duals of the generalised Gelfand–Graev characters and the vanishing assertion in Lusztig's theorem on unipotent supports [54, Theorem 11.2(iv)], again applicable by Taylor's Corollary 13.6, restrict the scalar products to the chosen rational classes. Their coefficients are the nonzero class sizes divided by the group order. If a multiplicity column were zero, duality would make the corresponding restricted dual orthogonal to the restriction of every ordinary character in the selected family. These restrictions span all functions on the classes, so that restricted dual would be zero, a contradiction.

The row sum is a separate deduction. The scalar product formula of Lusztig and Geck–Malle, in Taylor's form [70, Proposition 15.4 and Lemma 14.15], has right side $|\bar{A}|/n_{\rho^*}$ for $\rho \in \tilde{\mathcal{F}}'$. Its weights are the subgroup indices $[\bar{A} : \bar{A}^F]$. The component Frobenius is trivial, so these indices equal one. Under the Weyl Frobenius hypothesis, $n_{\rho} = |\mathcal{G}_{\mathcal{F}'}|$ [41, (6.1)(c)]. Alvis–Curtis duality preserves the denominator of the degree polynomial [45, Proposition 3.4.21], so $n_{\rho^*} = n_{\rho}$. Thus each row has sum one. Nonnegative integral entries and a nonzero entry in every column give an injective selection of the required characters. The denominator identity and the Weyl and component Frobenius identities are distinct external inputs.

Restriction to G is multiplicity free [45, Theorem 1.7.15]. The upper generalised Gelfand–Graev character at the same rational representative is induced from the lower generalised Gelfand–Graev character [70, proof of Lemma 14.12]. Alvis–Curtis duality commutes with restriction, with compatible signs for the positive duals [68, Lemma 5.2 and Corollary 5.3]. Frobenius reciprocity then expresses an upper scalar product as the sum of the lower scalar products over restriction constituents. This identity alone does not assert that the sum is zero or one.

If $Z_A = 1$, use the characters selected from $\tilde{\mathcal{F}}'$. The rational class map is injective, and ambient conjugation makes the entries over a singleton class equal. Their sum is one, so the restriction is irreducible and gives the required lower row. This includes the abelian cases with trivial central image and the corresponding case in rank three.

If $A = Z_A \cong C_2$, use the distinguished character from Taylor's original family [68, Theorem 3.2]. Its dual denominator is one and its restriction has exactly two constituents. Only the row of this distinguished character is needed from that family. Its scalar products with the upper generalised Gelfand–Graev characters sum to one by the same weighted formula. There is one upper class and two lower classes. Clifford transitivity exchanges the two constituents. Equivariance and nonnegative column sums of one force the resulting two by two matrix to be a permutation matrix.

For nonabelian A , keep Taylor's original distinguished character, whose restriction has two constituents. Its row sum is one, using its stated support and dual denominator. Over the singleton lower fibres use $\tilde{\mathcal{F}}'$. The distinguished character must lie above the double fibre. An element interchanging its constituents also interchanges those two classes, since otherwise two equal nonnegative entries would sum to one. The local matrix has the form

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix}, \quad a, b \in \mathbb{Z}_{\geq 0}, \quad a + b = 1,$$

and is a permutation matrix. A zero upper entry forces every lower entry over it to vanish. The original distinguished character and the family $\tilde{\mathcal{F}}'$ retain their own rational series labels. Unipotent supports are constant on each family [44, Theorem 3.7] and are preserved under restriction [70, Lemmas 14.12 and 14.15]. These statements apply to the characters chosen from either family. Their semisimple parameters are considered separately in the adjoint dual group. Taylor’s original parameter has quasi-isolated image, and the isolated second parameter also has quasi-isolated image [10, Proposition 2.3(b)]. Both images have order a power of two by Proposition 5.3(a) of that reference. The principal series identification therefore places the selected characters and their duals in the same principal block. The two families need not belong to one rational Lusztig series. Taylor’s identification of wave front sets with dual unipotent supports [70, Lemma 14.15], together with the support theorem of Achar and Aubert in the form stated in [70, Proposition 15.2], makes the full matrix triangular. Its nonsingularity proves independence of the P_u .

The counting assumption is the equality of the number of rational unipotent classes with $|\mathrm{IBr}(B_0(G))|$, from Chanab’s Proposition 2.7 and the principal series identification [24]. Projectivity comes from induction from subgroups of odd order and the specified modular reduction. The dimension of the projective character space follows from the decomposition data. Thus the P_u form a $\mathbb{Q}$ -basis of that space. This is a rational span of K -valued class functions. It does not assert that every value of Γ_u lies in $\mathbb{Q}$. The ordinary field K is the fraction field of a modular system with the required roots of unity. Algebraic closure of K is not an assumption.

Field automorphisms fix the P_u , so they fix their span and each projective indecomposable character. Duality then fixes every $\varphi \in \mathrm{IBr}(B_0(G))$. Its stabiliser in the special Clifford and field semidirect group consequently factors, and the cyclic extension argument supplies an irreducible extension to the field semidirect product. These conclusions include $(n, q) = (3, 3)$. We apply Feng–Yu–Zhang’s Theorem 5.16 [36] only in the nonexceptional case, with the required coefficient normalisation. The case $(n, q) = (3, 3)$ is treated separately below. The application also uses Späth’s comparison with the complete character triple condition [66, Theorem 4.4].

Implementation.

`ManuscriptIBAW.TypeB.GGGRSources` states assumptions on the ordinary characters, class representatives and coefficient field. In `FamilySource`, the weighted scalar product formula and the dual denominator equality are assumed separately. The component Frobenius is assumed to act trivially, and each weight is assumed to be the index of the corresponding fixed subgroup. Lean deduces that these indices are one and hence that each row has sum one. The identification of this action with geometric Frobenius remains external.

`IsolatedValueSource` assumes the trace expansions and diagonal pairings used in manuscript Lemma 2.11, together with the support and duality identities. The geometric argument establishing the trace expansions in the present setting of odd characteristic is the manuscript’s own argument and is not formalised. The stated Weyl group and its trivial Frobenius action are not used in the finite proofs. They have no formal identification with the Weyl group of the geometric centraliser relative to the chosen torus, and the denominator equality remains a separate assumption. The Borel subgroup, torus and character sheaves are not constructed here.

The trace expansions use inverse class representatives, as required by the K -valued scalar product. Constancy of ordinary characters on these inverse conjugacy classes is proved from conjugacy invariance. From the assumed expansions and diagonal pairings, `IsolatedValueSource.value_span` proves that the ordinary character values span all functions on the $\tilde{G}$ -classes, and `dualRestriction_independent` proves that the restrictions of the Alvis–Curtis duals of the generalised Gelfand–Graev characters are linearly independent. The theorem `localized_pairing` uses constancy on conjugacy classes and the support assumptions to express the pairing as a sum over those classes. The theorem `coverage` combines this identity with the assumed duality identity and proves that every multiplicity column has a nonzero entry. These

deductions use `ManuscriptIBAW.Library.FiniteCharacterSelection` and `ManuscriptIBAW.Library.FiniteClassPairing`, described in the library manual. Thus `NonabelianSource.isolatedCoverage` is proved from the stated assumptions.

`ManuscriptIBAW.TypeB.GGGRAbelian` derives `upper_delta_selection`, `singleton_rows` and `double_rows` from the same upper functions, actual restriction constituents and conjugation action. Its `rows` construction handles the two abelian central-image cases, including the rank three application. It does not assume the desired lower scalar product matrix. The structural descriptions of the component group and rational class map remain inputs. In particular, the surjective component map and its kernel identify the upper component group with A/Z_A . The implemented singleton argument selects a constituent with entry one and proves the other entries zero, which suffices for the matrix construction. `GGGRBasis.rankThreeSource` constructs the basis input `RankThreeChanebSource` from these proved rows.

`ManuscriptIBAW.TypeB.GGGRSelection` constructs `workingFamily` from the original distinguished character and the isolated characters over singleton fibres. The theorems `distinguishedParent_eq_original` and `singletonParent_mem_isolated` identify the characters whose restrictions contain the selected constituents. `SelectedClass.label` and `rowLabel` assign to each selected constituent the rational series label of that character. `ManuscriptIBAW.TypeB.GGGRBasis` uses these labels to prove `gggr_basis`, `principalBrauer_fixed` and `principal_selector`, with the abelian rows constructed by `GGGRAbelian`.

`ManuscriptIBAW.TypeB.PrincipalApplication` forms its `Inputs` from `ContextData`, the principal series identification, `GGGRSources.RankSources` and the field sources. Its public basis, fixedness and selector theorems apply `GGGRBasis`. For a relative family, `PrincipalRemainder` states the separate published orthogonal and central quotient hypotheses on `SpinGGGRInputs`. `PrincipalApplicationInputs.exists_witness` supplies the proved fixedness directly to `FYZ516SpathSource.apply516`, then applies the central quotient correspondence. The block, root, family and actions remain the same. The Levi argument uses this result on each spin factor.

4.8 The exceptional cover for $\ell = 2$

For manuscript Proposition 4.11, let $S = \Omega_7(3)$ and $X = 3.\Omega_7(3)$. The group used in Lean is constructed from a free central cover of the matrix group by quotienting its intrinsic central subgroup of order two. The structural sources identify the matrix group as nonabelian simple, identify the kernel of the full cover as cyclic of order six, and supply the universal central extension theorem for the free presentation. The construction proves that X is finite and perfect and that its projection $\pi : X \rightarrow S$ has kernel equal to $Z(X)$, of order three.

Its universal $2'$ -property is also deduced. The full cover maps to every perfect central extension with odd kernel. Its central subgroup of order two maps trivially, so the map factors through this same X . The resulting map remains surjective. Thus the cover used in the character argument has the required maximality, conditional on the identifications of the matrix group and free presentation.

The nine table labels $B_1, \dots, B_9$ are identified with all primitive blocks of X . For the principal block, the Brauer restriction calculation in `o7brauer.g` gives twelve Brauer characters in the principal block of S , with eight fixed by the noninner involution. Specialising the covering formulas of Feng–Yu–Zhang and identifying their weights with the specified group and block gives ten $\mathrm{SO}_7(3)$ -classes covering twelve S -classes, with eight singleton fibres and two double fibres [36, Lemma 5.7 and the proof of Proposition 5.11]. The covering relation identifies the action in each double fibre with that same involution [36, Corollary 2.22(2)]. The passage from $\mathrm{Spin}_7(3)$ to S uses the central 2-quotient correspondences for the principal blocks and their weights. Matching the involution's fixed points and orbits of size two constructs the

equivariant bijection. The central $2'$ -quotient argument lifts it through π using the same principal block and character deflations [30, Lemma 2.5].

For B_2 , the table calculation gives a defect group of order 8, five ordinary characters and two Brauer characters. One ordinary character has height 1, so Kessar–Malle’s theorem [48, Theorem 1.1] excludes abelian defect groups. The defect group of B_2 is therefore dihedral or quaternion. A block with quaternion defect group of order 8 has one or three irreducible Brauer characters [62, Theorem 8.1]. Since B_2 has two, its defect group is D_8 . The relevant result for dihedral blocks gives two weight classes, with radical orders eight and four [61, proof of Theorem 4.1]. Their different subgroup orders force both classes to be fixed by automorphisms, and the specified table action fixes the two Brauer characters. This gives the required equivariant matching.

Implementation.

`ManuscriptIBAW.TypeB.DefectEight` proves `defect_is_dihedral` from the table data and its count of two Brauer characters, the classification of groups of order eight, Kessar–Malle’s implication for abelian defect groups and the quaternion exclusion stated above. The block, defect subgroup, characters and roots are the same throughout, and these inputs are explicit assumptions.

`QuaternionExclusionSource.toNumerical` derives, by contradiction, the numerical implication used by the block construction. Both the ambient exceptional inputs and those for principal Levi factors require the quaternion exclusion itself. They derive the numerical implication before applying the block construction. The separate theorem `defect_is_dihedral` gives the direct dihedral argument.

The block B_3 has a defect group of order two and is nilpotent. The proof uses the local nilpotent argument underlying the criterion [50, Theorem 1.3]: it uses a defect representative of that same block, its Brauer correspondent and the unique ordinary character of defect zero on the local quotient. It proves that all radical subgroups occurring in weights of the block are conjugate to the defect group and proves that the block contains exactly one weight class. For the defect zero blocks B_4 and B_5 , the dominated blocks and their singleton Brauer fibres are also obtained before the local reductions and complete ambient and local block decompositions with their central characters are supplied.

Assume that the computed list contains every weight class with the given faithful central character. The list has eight classes in each faithful central sector. The blocks B_8 and B_9 have dihedral defect [55, Section 5.5]. The calculation in `o7blocks.g` gives two irreducible Brauer characters in each. Sambale’s theorem therefore gives two weight classes in each [61, Theorem 4.1], using the specified defect groups and all weight classes belonging to these blocks. Subtracting these weights leaves six in each of B_6 and B_7 . The noninner action swaps the two blocks in each pair, so each faithful block stabiliser consists of inner automorphisms. These fix both Brauer characters and weight conjugacy classes. The matching is therefore equivariant. The cyclic outer automorphism criterion then gives the inductive BAW conclusion, including its local and extension clauses [30, Corollary 2.13].

The coefficients in this exceptional application come from one specified modular system. Its fraction field K has the required finite splitting roots for X , with a separate assumption on roots of unity for $\mathrm{SO}_7(3)$ in the principal source. Only the residue field is required to be algebraically closed. The root embedding for S is derived from the one for X and retains its residue compatibility. Every local reduction concerns the weight’s own ordinary character on its own normaliser quotient. Local extension values are equated with that same local character.

The blockwise arguments use the natural projection π for B_1 – B_5 and the identity on X for B_6 – B_9 . They retain the block images, character kernels, deflation values and quotient identifications on these presentations. A common family is then formed on $X = 3.\Omega_7(3)$, using the same modular system, root correspondence and local block operations.

Two separate assumptions enter the passage to the full condition. The first is an additional assumption for these fixed groups and coefficients. It interprets the natural quotient criteria as the relative block conditions on this common family. It includes the passage from the full cover $6.\Omega_7(3)$, character and block identifications over the fixed coefficient fields, and compatible root correspondences for quotient groups, extension groups and intermediate subgroups. Späth's Theorem 4.4 concerns the character triple and extension part of this passage [66]. The second assumption combines the relative block conditions into a compatible family, with the normalisation required by the full inductive condition. The proof of the full family condition uses the quotient transport and the passage to compatible correspondences as separate assumptions.

Implementation.

`ManuscriptIBAW.TypeB.ExceptionalTarget` defines the cover and `Target.Complete` independently of the block proof. `ManuscriptIBAW.TypeB.ExceptionalFamily` constructs `commonFamily` and `commonCover`, including the selected character reductions and their root identities. The additional `AmbientIdempotents` premise identifies every block idempotent with the one used in the local block operations. In `ManuscriptIBAW.TypeB.ExceptionalApplication`, `Inputs.target` fixes the natural target from `raw`, and `Inputs.completeNatural` proves the blockwise conclusion. `Inputs.familyComplete` applies the explicit `NaturalQuotientTransport` for these fixed data and the separate `Inputs.compatibleFamilySource` for the passage to the full condition. `Inputs.complete` then constructs `FamilyCertificate`, containing that same family, cover and witness. The equivalence between the nine labels and all blocks is proved from the specified data. The table, group, character, block and coefficient identifications remain assumptions.

The principal quotient application uses

`ManuscriptIBAW.TypeB.ExceptionalPrincipalDescent`. Its `PrincipalSources.prop411` is one record of the exceptional group hypotheses. The theorem `Inputs.familyComplete` constructs the full family from that record. The character and weight inflation maps in `BlockTransport` then transport its principal block bijection to $\Omega_7(3)$. Lean proves that this map is equivariant. The `Source` in `ModularRep.CurrentSpathFiniteSplittingDescent` assumes descent of the extension and block conditions for every pair in this specified transported bijection. The application uses [65, Proposition 4.6] and [66, Theorem 4.4] for this passage. The formal source requires the stated central quotient, character interpretations, full block stabilisers and compatible coefficient fields. `BlockTransport.downAutomorphisms` identifies the acting group on the quotient with its full block stabiliser, while the covering family includes the corresponding identification on the cover. No instance of this descent source is constructed here. Both ordinary fields must contain the roots required by the extension groups in the covering family. Their being fraction fields does not imply algebraic closure. The theorem `PrincipalSources.coherent` therefore uses the family already proved from `prop411`. No independently assumed principal block matching is used in this application.

The radical computation examines all 3021 conjugacy classes of subgroups of a Sylow subgroup and finds 26 radical representatives, which form twelve conjugacy classes in X . It counts ordinary defect zero characters of their normaliser quotients by central sector. The interpretation in Lean must identify these outputs with the specified matrix and `AtlasRep` groups, blocks and roots. The two faithful central characters may be interchanged when the computations use different generators of the centre. The two local principal weights do not themselves establish the separate global count of ten orthogonal principal weight classes.

4.9 Nonprincipal representatives and Jordan reduction

The supporting lemmas and the principal result give the full Brauer hypothesis in manuscript Proposition 4.13. Let b be a nonprincipal block of $G = \text{Spin}_{2n+1}(q)$. Its semisimple parameter s is considered up to rational conjugacy in the projective conformal symplectic dual group, with series idempotent e_s . The rational class is assumed to be characterised uniquely by the block support equation $be_s = b$. Diagonal automorphisms fix these group algebra idempotents

[11, Lemma 7.4], while field automorphisms permute them according to their semisimple parameters [71, Proposition 7.2]. These identities are assumed on the same group algebra idempotents.

For this parameter, choose a proper Levi subgroup dual to the unique minimal Levi containing the connected and rational centralisers, after a common rational conjugation. The existence and properties of this choice are geometric assumptions. The simultaneous stability of the paired Levi subgroup and its idempotent under the chosen Frobenius power is the assertion used from [60, Lemma 4.5(a)–(b)]. The field group used here is the stabiliser of the same parameter. The supplied Jordan equivalence is defined on the entire original Brauer sets belonging to these idempotents. Its block comparison retains the equation for the supporting block [31, Lemma 5.1 and the proof of Proposition 5.2]. These geometric and character identifications remain external.

A block of the derived Levi group covered by a block belonging to e_s is strictly quasi-isolated, as in the proof of [31, Theorem 5.7]. Its blocks on the simple factors are strictly quasi-isolated by the proof of Proposition 5.6 of the same paper. On the Type B factors these blocks are principal by the principal series argument above. These block and parameter identifications are included in the factor assumptions.

For Type A, the representative theorem [27, Theorem 8.1] includes degree two and $\mathrm{SL}_2(3)$. For a rank two factor $\mathrm{Sp}_4(q_0)$, trivial field action gives the factorisation directly. Otherwise the source requires $q_0 \geq 9$ and the stabiliser assertion for its principal 2-block from the verification following [18, Lemma 4.5, p. 1203], with its block condition in Lemma 4.6 of the same paper. Diagonal conjugation gives this assertion at the chosen principal character. Its interpretation on the matrix group and specified roots remains an assumption. Every spin factor of higher rank supplies the geometric data and principal block assumptions stated above. The principal series identification is proved on these data before applying the factor theorem. In particular, lower factors over the field of order three are included even when the ambient pair is nonexceptional.

The component and Clifford arguments construct a Levi representative, using one conjugating element for its character and chosen constituent. The rational product and Lang argument also give the ambient product with the paired Levi. The injective Jordan map transfers the factorisation to the original character. Uniqueness of its block parameter then enlarges the factorisation from the parameter stabiliser to the full field group. Cyclic extension is applied to the original irreducible representation. The proof handles the prime field case and both principal and nonprincipal characters, so it supplies a representative in every original special Clifford orbit.

For each semisimple label, fix a group with the required outer quotient and image of the series stabiliser, using the geometric choice in [60, Corollary 4.2]. The geometric assumptions also specify the Frobenius powers and commutation relations, and the stability of the Levi and rational class. This group is fixed for all relevant character orbits. The factorisation and extension proved for the full field group restrict to the required condition for the chosen group. The character and its affording representation remain the same under this restriction. This is the first hypothesis of the reduction, as used in manuscript Lemma 4.12.

The second hypothesis ranges over the complete relative class of algebraic pairs in Feng–Li–Zhang’s Hypothesis 5.5 [31], including the ambient pair. Type A uses the stated application to all blocks [32], with its hypotheses for small groups [29]. For a relative Type B group, the strictly quasi-isolated block is principal. For a relative factor of rank two over an odd field, the manuscript uses $\mathrm{Spin}_5 \cong \mathrm{Sp}_4$ and the principal block bijection constructed at the start of Proposition 3.3. The original relative character triple relation and coefficient conventions are preserved. Higher spin groups use the principal result and the passage through their central 2-core. The lower exceptional group uses the nine-block theorem on $3.\Omega_7(3)$, followed by fixed

point descent and the lift to the original spin block. No relative pair is discarded because it has a central 2-subgroup or is not itself a universal 2'-cover.

The geometric labels in these arguments are identified with the original block parameters. The ambient series idempotent is the sum of the primitive idempotents over its rational dual conjugacy class, and equals the idempotent in the parameter source. The finite dual is identified with the specified projective conformal symplectic group, its geometric parameter represents that same class, and its field action agrees with the original rational action. Orthogonality and the two support equations prove that the geometric class of each block equals its original block parameter. Each application to a rank two or spin factor also identifies the parameter used for strict quasi-isolation with the parameter used to prove principality. Jordan reduction uses these equalities. The interpretation as algebraic semisimple elements and rational Lusztig series remains an assumption.

For the nonexceptional ambient pair, the final family is the primitive family on $S = \Omega_{2n+1}(q)$. Its covering map is the identity on S . This universal 2'-cover is constructed from the spin projection, its centre of order two and the stated full universal covering property. Here the condition $(n, q) \neq (3, 3)$ concerns only the ambient pair. The exceptional ambient pair uses the separate argument for the triple cover.

Feng–Li–Zhang’s Theorem 5.7 gives BAW-goodness once its two hypotheses are established [31]. To obtain the full inductive condition for the blocks of S , the argument also assumes the passage through the central 2-core for blocks and weights, together with Späth’s comparison with that condition [66, Theorem 4.4]. Every block of S is assumed to have a dominating spin block under the original group algebra map. Deflation uses the same modular system, roots, local characters and extensions. A further hypothesis supplies compatible choices of block bijections and extension data over all blocks of S , in the prescribed modular system and with the same roots, block operations and local reductions.

For the ambient group and spin factors of higher rank, the ordinary field is the fraction field of a modular system with the required finite roots. The algebraically closed ordinary fields permitted by the relative Type A and rank two sources are confined to those families. They are not imposed on the spin modular system.

Implementation.

In `ManuscriptIBAW.TypeB.LeviRankTwo`, `Sources` specifies the original principal block, a character belonging to that block and the geometric identification with Sp_4 . Its character assumption allows either trivial field action on the group or the principal block assertion over a field of order at least nine. Triviality here is a hypothesis about the action on the group. The proof of `standard_factorization_at` uses separation only at the chosen character. The theorem `factorization_on_original` transports the factorisation through the same group and character identifications. Thus `Sources.representative` gives a representative on the original Levi factor without a premise about all Brauer characters of Sp_4 . In `ManuscriptIBAW.TypeB.LeviSources`, `SpinFactorSources` contains `PrincipalSeries.ContextData` and `PrincipalApplication.Inputs`. `SpinFactorSources.representative` applies the principal fixedness theorem on that factor’s groups, characters and roots. The theorems `factor_representatives`, `selected_constituent` and `same_y_representative` combine the factor and component arguments. In the same namespace, `ManuscriptIBAW.TypeB.LeviApplication` proves `Source.selected_jordan` and `brauerHypothesis`, retaining the Jordan equivalence, block correspondence and equivariance equations. `TwoApplication.Inputs.brauerHypothesis` applies this result, and `Inputs.perLabel` restricts it after the geometric groups have been selected. In Lean, `JordanRouting.rankTwoBijection` works in a splitting modular system and gives an iBAW bijection for the original relative block by lifting the Li–Li quotient block bijection. The covering maps, block identifications, quotient and root interpretations and character triple conditions for this relative factor remain explicit. `JordanStrictApplication.BranchSources.blockWitness` proves the

relative block condition for each Type A, rank two or higher spin case.

`JordanStrictApplication.Sources.strictBlocks` combines these conditions over the relative strictly quasi-isolated blocks. The records `JordanRouting.FullLabelBinding` and `Jordan.TypeBLocalized.AmbientSeriesBinding` keep the strict labels and ambient rational parameters identified. `AmbientSeriesBinding.block_parameter_class` is a deduction. Finally, in `ManuscriptIBAW.TypeB.TwoApplication`, `Inputs.ggg` contains the geometric and character data, and the Brauer argument uses `Inputs.principal.context` after construction. `Inputs.family`, `Inputs.cover` and `Inputs.target` are defined before `Inputs.complete` establishes the full inductive condition at the nonexceptional pair.

4.10 All primes, small ranks and even characteristic

For odd q and $n \geq 3$, consider a prime divisor ℓ of $|S|$. If $\ell = p$, apply the defining characteristic result. If ℓ is odd and nondefining, apply the result for odd primes. For $\ell = 2$, use the generic result when $(n, q) \neq (3, 3)$ and the result for the exceptional triple cover at $(n, q) = (3, 3)$. Every case concerns the same matrix group S . It supplies the complete condition on its specified cover, retaining the specified family, projection, characters and roots. In particular, the exceptional case for $\ell = 2$ uses the common family obtained from the argument for the nine blocks under the separate assumptions on quotient transport and passage to compatible correspondences.

Each case uses the structural and published assumptions already stated in its proof. The complete condition follows from these assumptions. Applying the argument to every prime divisor of $|S|$ gives the result for all primes in odd characteristic and rank at least three.

Implementation.

`ManuscriptIBAW.TypeB.AllPrimes` defines the assumptions for these cases in `Inputs`. Its `complete` and `all_primes` apply the preceding proofs on the same orthogonal group. `OddHighInputs.complete` combines these with the common structural assumptions on the finite field and Clifford projection.

To cover all ranks and fields, distinguish rank one, rank two over $\mathbb{F}_2$, the cases identified with symplectic groups, and odd characteristic in rank at least three. The group isomorphisms are structural assumptions. Rank one uses the specified $\mathrm{PSL}_2(q)$ identification and the published low-rank result [29, Proposition 4.6]. The derived-group convention at rank two over $\mathbb{F}_2$ uses $\mathrm{PSL}_2(9)$. The low-rank source includes a cover of a nonabelian simple target, so it is not applied to the nonsimple groups $\mathrm{PSL}_2(2)$ or $\mathrm{PSL}_2(3)$.

The other rank two cases and the cases in even characteristic use the Type C theorem at the same rank, field and prime. These cases satisfy $(n, q) \neq (2, 2)$, as required by that theorem. Composing the covering map with the specified isomorphism of simple groups retains the covering group, complete family, characters and roots.

Implementation.

`ManuscriptIBAW.TypeB.Census.complete_at_case` performs this case distinction. The `AllPrimeFamily` result for the same orthogonal group in higher rank is proved by the preceding distinction among primes and supplied separately from `CaseInputs.Census.of_presentation` transports the result along the stated Type B presentation. Finally, `ManuscriptIBAW.TypeB.finite_simple_typeB` combines this case distinction with `OddHighInputs.complete` to prove the condition for every nonabelian simple group with the specified Type B presentation, under its stated source assumptions.

Chapter 5

The sporadic arguments

Let S be a sporadic simple group, let ℓ divide $|S|$, and write $X_\ell(S)$ for its specified universal ℓ' -cover. The manuscript treats the 45 relevant pairs for J_4 , F'_{24} , the Baby Monster and the Monster, then combines them with the verification reported in the CTBlocks overview for the other 22 groups.

The arguments use numerical weight equalities, the action of the outer automorphism group, and calculations with ordinary character tables. Their hypotheses specify the covering groups, blocks, local characters and the interpretation of the table entries. The local extension and block compatibility conditions are also required for the full inductive conclusion. The ordinary values of Brauer characters depend on the chosen identification of roots of unity. Whenever a global character, a local character or their restrictions to an intermediate group are compared, these identifications must agree on the roots involved. The complete certificates require this agreement for their extensions. For the direct constructions, root agreement is assumed for the extensions already constructed. For the triple cover, a separate assumption supplies replacement extensions and intermediate block witnesses with compatible roots for each pair in the same numerical matching. Their existence is not derived from the extension and block assumptions used in the original construction. A certificate formed without the required root compatibility is retained only as a `RawDefinition41Certificate` and does not establish the complete inductive condition.

5.1 Two general arguments

Suppose first that $\text{Out}(X_\ell(S)) = 1$ and that

$$|\text{IBr}(b)| = |\text{Alp}(b)|$$

for each block b of $X_\ell(S)$. Choose a bijection on each block. Every automorphism of the cover is inner, so it fixes each irreducible Brauer character and each conjugacy class of weights. The chosen bijections are therefore equivariant under the block stabilisers. The identification of the covering group also gives $\text{Out}(S) \cong \text{Out}(X_\ell(S))$.

The cyclic outer criterion in [30, Corollary 2.13] gives the inductive BAW conclusion for each block. The construction chooses the block bijections with the prescribed defect zero normalisation and proves the required extension and block conditions from the local reduction and quotient laws, with the compatible root choices specified below. It uses the original cover, including a nontrivial centre. This proves the implication in manuscript Lemma 5.2.

The second argument is manuscript Lemma 5.3. A finite set with an involution is determined up to equivariant bijection by its total size and number of fixed points. These two numbers add

over invariant disjoint unions. Thus an equivariant bijection on the whole set, together with equivariant bijections on all but one block orbit, gives an equivariant bijection on the remaining orbit by subtraction. If that orbit consists of two exchanged blocks, choose a bijection on one block and define the other by the involutions. This ensures that the resulting map preserves individual blocks. The library manual gives the complete proof for finite sets.

Implementation.

`ManuscriptIBAW.Sporadic.CompleteCollapse` states the numerical premise as `NumericalHypothesis` on the actual blocks of a fixed `CaseModel`. Its `Sources` contains local reductions, the laws for defect zero blocks and trivial weights, local block compatibility, the quotient data and the prescribed coefficient field. The quotient data specify finite complete block decompositions for the quotient group and its relevant normalisers, together with their block central characters, before a correspondence is chosen. They also supply the required normaliser quotient and block induction laws. `lemma_5_2` constructs the complete `CaseConclusion` from these assumptions, the numerical premise and the fact that all automorphisms are inner. The ordinary field has characteristic zero and the modular field is algebraically closed. The character interpretations are supplied for these fields. The full `Definition41Certificate` also requires `OriginalWitnessRoots` or `CanonicalWitnessRoots` for the extensions in the constructed witness. For the centreless argument, the corresponding source is `CoherentOriginalRowsOutput`. For the quotient, extension and intermediate block witnesses recorded in an original packet, `OriginalPacketRoots` requires agreement between the original and quotient roots, between the global extension and quotient roots, between the global and local extension roots, and with both root correspondences used on J and $D \cap J$ for every intermediate subgroup J , where D is the normaliser used for the local extension. Agreement between the global extension and all original roots is an additional condition in `ActualPacketRoots`, used by `CanonicalWitnessRoots`. It is not required for an original packet, whose base representation factors through the central character quotient. The quotient roots agree with the original Brauer character roots. The inflated quotient representation and the restricted global representation therefore afford that same Brauer character. `RawDefinition41Certificate` omits these root compatibility conditions and is used only in intermediate constructions. `CaseConclusion` uses `Definition41Certificate`, which includes them. For the triple cover, `CoherentOriginalPacketChoices.OriginalPacketChoices` requires compatible extension and block witnesses for every pair of one specified bijection. The applications at primes two, five and seven and the applications to cyclic blocks apply this assumption only to the bijection selected by their numerical argument. They do not assume it for every equivariant block bijection. The constructor `withCompatiblePackets` preserves that bijection, its equivariance, its block correspondence and its defect zero normalisation. It replaces the packet field of the constructed witness with packets having compatible roots supplied by `OriginalPacketChoices`. The original packets satisfy the raw restriction and intermediate block conditions, but their root compatibility has not been proved. For the replacement packets, `OriginalWitnessRoots` follows directly from the assumption, rather than from the raw packet construction. The existence of compatible extension and block witnesses for the numerical matching therefore remains an explicit assumption. `ManuscriptIBAW.Sporadic.Cancellation` gives the finite set deductions, including `cancel_exchanged_blocks_preserving`.

5.2 The covers and the group J_4

The multiplier and outer group data are those of the Atlas [26, Table 1]. The covers are

S	$X_\ell(S)$
J_4	J_4
Fi'_{24}	Fi'_{24} at 3, $3.Fi'_{24}$ otherwise
B	B at 2, $2.B$ otherwise
M	M.

Here B and M denote the Baby Monster and the Monster. Each group name is tied to its covering projection and specified simple quotient. Those identifications are part of the structural assumptions.

For J_4 , the blockwise weight counts for noncyclic defect are An–O’Brien–Wilson’s Theorem 5.2 [7]. Späth’s Proposition 6.2 supplies the cyclic cases, including defect zero [65]. The multiplier and outer group are trivial, so the first general argument applies at every prime dividing the group order.

Implementation.

`ManuscriptIBAW.Sporadic.J4` records three clauses in `NumericalSources`: existence of a defect group for each actual block, Späth’s equality for cyclic defect, and An–O’Brien–Wilson’s equality for noncyclic defect. `NumericalSources.all_blocks` proves the equality for every block by that case distinction. The structure `Inputs` keeps the selected finite group, roots, actual block operations, local reductions, full cover and outer group data, and its identification with the named simple base. `Inputs.model` fixes the required cover before `Inputs.complete` gives its complete inductive certificate. The ordinary field has characteristic zero and the modular field is algebraically closed of characteristic ℓ . The published numerical equalities require the stated splitting and character interpretation. The group identification is an additional structural assumption.

5.3 The central sectors of the Fischer cover

At $\ell \neq 3$, set $X = 3.Fi'_{24}$ and $Z = Z(X)$, and let $\pi : X \rightarrow \bar{X} := X/Z$ be the quotient map. A bar denotes the image of an element or subgroup under π . For an ℓ -subgroup $\bar{Q}$ of $\bar{X}$, the inverse image $\pi^{-1}(\bar{Q})$ has a complement Q to Z . Schur–Zassenhaus gives existence, and centrality of Z makes this complement unique. Radicality is preserved by this correspondence [59, Lemma 2.3(c)]. The quotient map gives

$$\overline{N_X(Q)} = N_{\bar{X}}(\bar{Q}), \quad N_X(Q)/(QZ) \cong N_{\bar{X}}(\bar{Q})/\bar{Q}.$$

Inflation identifies the blocks and Brauer characters of $\bar{X}$ with those of X over the trivial character of Z (see, for example, [58, Theorem 9.9(c)]). The corresponding weight character is inflated along

$$\pi_Q : N_X(Q)/Q \rightarrow N_{\bar{X}}(\bar{Q})/\bar{Q}, \quad nQ \mapsto n\bar{Q}.$$

Here $n\bar{Q}$ denotes the image under this quotient homomorphism. The block correspondence and the equivariant weight passage use Feng–Li–Zhang’s Lemmas 2.3(iii) and 2.5 [30], with the same local character and block operations.

By [5, Lemma 4.7], the outer involution exchanges the two faithful central characters and hence their sectors. A block in a faithful sector consequently has only inner automorphisms in its stabiliser. A bijection for one such block is equivariant under that stabiliser, and applying the outer involution gives the bijection on its partner.

For a finite set Y with an involution, write $\text{sig}(Y) = (|Y|, |Y^{C_2}|)$. Table 8 of An–Dietrich [5] gives the weight counts used below. Its entry a/b means $a + b$ weights, with a fixed by the involution.

5.4 The Fischer group for $\ell = 2$

The trivial sector has five blocks with pairs of total and fixed Brauer character counts

$$(33, 25), \quad (3, 1), \quad (3, 3), \quad (1, 1), \quad (1, 1).$$

These are obtained from the ordinary restrictions and outer action in the specified computation. The last two blocks have defect zero and use the bijection $\chi^0 \mapsto (1, \chi)$ with their unique ordinary irreducible character.

The two other nonprincipal blocks have defect groups of orders 4 and 8 [4, Lemma 4.2(d)]. The numerical Alperin weight equality for blocks of defect at most four [62, Theorem 13.7] therefore gives three weights in each. Their fixed counts follow from invariant subsets of the conjugacy classes of radical subgroups in that table.

The classes $(2^2)_a$ and $(2^2)_b$ in Table 8 are called 2^2 and $(2^2)^*$, respectively, in [4, Table 2]. For the block of defect two, every weight subgroup is conjugate into its defect group [58, Theorem 4.14]. The classification places that defect group in the class denoted $(2^2)_b$. The rows for the trivial subgroup account for the two defect zero blocks, and the row of order two contributes no weights to this sector. Thus the three weights form an invariant subset of the four weights in the $(2^2)_b$ row. This row has two fixed weights and one orbit of length two. An invariant subset of size three contains that orbit and one fixed weight. Its signature is therefore $(3, 1)$.

For the block with defect group D_8 , the possible conjugacy classes of radical subgroups are $(2^2)_a$, $(2^2)_b$ and D_8 . The first and last rows each contain one fixed weight. The middle row contains two fixed weights and one orbit of length two. Three of its four weights have already been assigned to the block of defect two. The only remaining candidates are the $(2^2)_a$ weight, the remaining fixed $(2^2)_b$ weight and the D_8 weight. There are three weights in the block, so all three occur and are fixed. Its signature is $(3, 3)$.

The total signature of the trivial sector is $(41, 31)$, from Table 8. Cancel the two small positive defect blocks and the two defect zero blocks. The principal signature is $(33, 25)$, equal to its Brauer signature. The argument for sets with an involution gives the required block bijections.

Each faithful sector contains exactly two blocks and has a total of 25 weights. The blocks have defects 21 and 3 and contain 23 and 2 Brauer characters, respectively. The block of defect three has two weights by Sambale's theorem, so the other block has 23 weights. Bijections on one faithful sector are transported to the other by the outer involution.

The fusion calculation for the $(2^2)_a$ and $(2^2)_b$ rows gives the same block distribution as the invariant subset argument. Both calculations use the specified radical and character table data. Implementation.

`ManuscriptIBAW.Sporadic.Fi24TwoRows` identifies the finite sets of ordinary characters of defect zero of the normaliser quotients with the weight classes in the trivial sector, using the same central quotient, normaliser characters, reductions and root restrictions. Its `row_total` and `row_fixed` give their total and fixed counts. Compatibility of the automorphism actions under this identification also proves that each such set is invariant.

`ManuscriptIBAW.Sporadic.Fi24TwoSupport.Geometry` specifies the two actual defect groups and classifies the radical subgroups conjugate into them. These are subgroup assertions, without a local character or an assignment of weights to blocks. The Brauer support criterion proves that the subgroup of a weight is conjugate into the appropriate defect group. A weight with trivial radical would belong to a defect zero ordinary block with one Brauer character, so it cannot occur in either block with three Brauer characters. The row corresponding to a radical subgroup of order two has specified count zero and is therefore empty.

The resulting support statements place the weights of the first small block in the set of four weights associated with $(2^2)_b$ and those of the second in the three possible nonempty sets. The chosen finite enumeration contains each radical conjugacy class exactly once. The sets of weight classes associated with $(2^2)_a$, $(2^2)_b$ and D_8 are disjoint. Their union contains six weight classes, four fixed by τ .

`Fi24TwoCounting.three_in_four` and `Fi24TwoCounting.remaining_three` prove the two subset calculations. The small defect theorem supplies the weight counts. Their fixed counts are conclusions of `Fi24TwoWeights.fixed_counts`.

`ManuscriptIBAW.Sporadic.Fi24Two.Inputs.model` fixes the groups, roots, local block operations and universal triple cover before `Inputs.complete` constructs the full condition. The numerical argument combines the derived signatures, cancellation and the defect zero normalisation to choose a correspondence. The additional `Inputs.packets` hypothesis supplies compatible extension and intermediate block witnesses for each pair of that correspondence. `Inputs.complete` uses those witnesses on the same model. The ordinary character values, block labels and outer class fusion are specified for these groups. The relevant quotients are also supplied with finite complete block decompositions and their block central characters. The subset argument supplies the assignment of the weights in the small blocks and their fixed counts. Existence of the compatible witnesses is a separate hypothesis and is not derived here from the counting argument or the raw packet construction.

5.5 The Fischer group at the other primes

For $\ell = 3$ the cover is $G = Fi'_{24}$. The classification supplies a radical subgroup $Q \cong C_3^2$ with

$$N_G(Q) \cong (3^2:2 \times G_2(3)).2$$

[4, Proposition 4.1 and Table 1]. The same source gives the unique nonprincipal block B_1 of positive defect, its defect group Q , and four weights [4, Lemma 4.2(c) and Theorem 4.3].

The character table calculation selects the four ordinary defect zero characters of $N_G(Q)/Q$ and inflates them to the specified local block. Brauer block induction is defined by the criterion for defect groups. The reduced central characters of induced ordinary characters identify the induced block of G [58, Theorem 4.14 and Lemma 6.1]. The calculation tests each entry of the list of 32 compatible class fusions. For every candidate and each of the six ordinary characters in that local block, the test identifies block 2 of G . The defects of the blocks of G are 16, 2, 0, with the first block principal, so this second block is B_1 . The test uses all 32 entries of the candidate list, whether or not some entries coincide.

The four selected ordinary characters therefore account for all four weights of B_1 . The radical row in Table 8 has two fixed weights, giving signature (4, 2), equal to the Brauer signature. The defect zero block has its usual bijection with one weight. Cancellation from the equivariant sector correspondence of An–Dietrich [5, Section 4.3.4] gives the principal signature (25, 25) and a bijection preserving blocks.

Implementation.

`ManuscriptIBAW.Sporadic.Fi24ThreeLocal.NamedTableSource` specifies the normaliser block containing the four inflated ordinary characters and the identity of reduced central characters after the actual Brauer map. Evaluating this identity on the target primitive idempotent proves `row_block`. The interpretation of the named rows, actual inclusion and candidate fusions remains a composite source assumption. The separate published total of four, together with injectivity of the four local classes, proves `radical_support`.

`ManuscriptIBAW.Sporadic.Fi24Three.SourceInputs` also specifies the six ordinary rows of the block on its thirty regular classes, a common root encoding and the actual outer class fusion. The matrix certificates give rank four and symmetrised rank three. Under the stated decomposition and evaluation identifications, these imply four Brauer characters and two fixed characters. The local character set is identified with the four characters in the 3^2 row of [5, Table 8]. The field `rowBijective` identifies all four ordinary characters of defect zero of the actual normaliser quotient. The field `localFixed` takes the two fixed characters in the entry 2/2 as a published count for the specified local automorphism. These are explicit source assumptions. No unverified local fusion is used to deduce the fixed count.

`SourceInputs.complete` combines these deductions with the An–Dietrich correspondence on the whole sector and the canonical defect zero normalisation. Cancellation constructs the principal block matching. The full condition on the fixed `Inputs.model` additionally requires the stated compatible local extension and intermediate block witnesses for this same matching.

For $\ell = 5$, the nontrivial radical subgroups of Fi'_{24} have orders 5 and 25 [4, Proposition 4.1 and Table 1]. The argument uses their unique lifts to $3.Fi'_{24}$, the corresponding normaliser quotients and inflated characters. For every tested fusion, each selected character induces to a unique block, and the block distribution is independent of the candidate. The outer class fusion preserves the selected character rows and induces an involution on them. The three invariant block signatures in the trivial sector are

$$(16, 16), \quad (14, 6), \quad (16, 16).$$

They equal the corresponding pairs of total and fixed Brauer character counts. The two pairs of exchanged faithful blocks have respectively 16 and 14 weights and Brauer characters per block. The weights whose radical subgroups have order five belong to cyclic defect blocks, and the cited classification accounts for all nontrivial radical classes.

For $\ell = 7$, each sector has a unique block of noncyclic defect. Use the equivariant correspondence on the whole sector [5, Section 4.3.4 and Table 8]. Removing the cyclic and defect zero blocks gives signature $(22, 12)$ in the trivial sector. Each faithful sector has 63 weights in total. Removing its known blocks leaves 22 weights, equal to the Brauer count in the unique noncyclic block. Choose a bijection for one faithful block and transport it to the other.

At 11, 13, 17, 23, 29, the full defect distributions in the specified computation show that every positive defect block is cyclic. Späth's Proposition 6.2 treats these blocks and all the defect zero blocks, as well as the remaining cyclic blocks at the smaller primes [65]. This gives equivariant block bijections on every required Fischer cover. The outer group is cyclic of order two, so the manuscript invokes the cyclic outer criterion to finish the argument.

Implementation.

The namespaces `ManuscriptIBAW.Sporadic.Fi24Five`, `Fi24Seven` and `Fi24Cyclic` construct their `Inputs.model` directly from the specified groups, roots, local operations and covering projection, before proving `Inputs.complete`. For $\ell = 5$, the seven blocks in the calculation carry separate Brauer and weight counts, their fixed counts and the outer permutation. These equations and the condition that every block outside the seven specified blocks has cyclic defect are source assumptions. The finite comparison gives all the required block bijections.

For $\ell = 7$, the assumptions specify whole sector totals, the fixed total in the trivial sector and the condition that every block other than the distinguished block in each sector has cyclic defect. The equality for that remaining block is deduced by cancellation. At the five listed larger primes, the source supplies an actual cyclic defect group for every block. A defect exponent at most one suffices after the block and defect interpretation.

The full inductive conclusion for the original triple cover also uses the supplied compatible extension and intermediate block witnesses for each pair of the selected numerical correspondence. These witnesses concern the same normaliser characters, reductions and root correspondences. The modular field is algebraically closed of characteristic ℓ . The ordinary field has characteristic zero, with the compatible splitting and character interpretation required by the published data.

For the triple cover, the numerical arguments first construct a common correspondence with the defect zero normalisation. The full inductive condition then uses the compatible packets supplied for that correspondence, in addition to the ordinary and modular extension, covering and block laws. The centre action and map from the full cover then separate the faithful and trivial cases in the local proofs. For the centreless case with $\ell = 3$, the ordinary character family is combined with block preservation, defect zero normalisation and the full local data to give the complete inductive certificate.

Implementation.

`ManuscriptIBAW.Sporadic.CyclicOuterFamilies` provides `TripleCover.ofNormalizedEquiv` and `Centerless.centerless_manuscript_ad3_assembly`. The first constructs a raw witness using

the common normalised correspondence on the original triple cover. It does not establish the compatibility of roots required by the full certificate. The applications use `withCompatiblePackets` with the supplied compatible packets to obtain that certificate without changing the correspondence. The second constructs conditions (3.a)–(3.d) for AWC-goodness in Navarro and Tiep [59, §3.3], in the form recalled by An and Dietrich [5, Definition 4.4], before the full inductive condition is deduced. Both constructions use the stated outer group, centre and coefficient hypotheses.

5.6 The Baby Monster and the Monster

For the Baby Monster and $\ell = 2$, the ordinary table has two blocks of defects 41 and 3, containing 179 and 5 ordinary irreducible characters. The ordinary table is verified in [15, Section 7]. The block decomposition and the restriction ranks are separate table calculations. Restriction to the 2-regular classes has exact ranks 25 and 2. The restriction rank argument therefore gives 27 irreducible Brauer characters in total.

An–Dietrich compute 27 conjugacy classes of 2-weights [5, proof of Theorem 4.2 and Table 2]. The corrections to the subgroup descriptions in [6, Table 2] leave this total unchanged. The nonprincipal block has a defect group of order eight, so its number of weights equals its Brauer count, namely two, by [62, Theorem 13.7]. Subtracting gives 25 weights for the principal block. The outer group is trivial, so the first general argument gives the inductive conclusion on B.

At odd primes the cover is 2.B. The noncyclic defect equalities are An–Wilson’s Theorem 5.3 [8], and the cyclic cases use Späth’s Proposition 6.2 [65]. For $\ell = 7$, the proof of Lemma 5.2 in [8] lists 220 regular classes. The appendix calculation gives 222. The four relevant block restriction ranks are still 24, 24, 21, 24, so this correction does not change the blockwise values used in Theorem 5.3. The outer group of the double cover is again trivial.

For the Monster and $\ell = 2$, the ordinary table has five blocks, with defects

$$(46, 0, 0, 4, 0)$$

and ordinary character counts (183, 1, 1, 8, 1). The ordinary table verification in [16, Sections 2 and 5] assumes that there are precisely two conjugacy classes of involutions, with the specified centralisers, and that an ordinary irreducible character of degree 196883 exists. It also uses the group, subgroup and representation data set out in Section 2. The block distribution above is a separate calculation with the ordinary table. The table calculation gives 61 conjugacy classes of 2-regular elements. Hence there are 61 irreducible 2-Brauer characters (see, for example, [58, Corollary 2.10]). An–Dietrich compute 61 conjugacy classes of 2-weights [5, proof of Theorem 4.2 and Table 17]. The corrections to the subgroup and normaliser entries in [6, Table 17] leave this total unchanged.

The three defect zero blocks each contribute one character and one weight. The remaining nonprincipal block has a defect group of order sixteen, so its weight count equals its Brauer count by [62, Theorem 13.7]. These five blocks form the complete block decomposition, so they account for every nonprincipal contribution. Subtraction therefore proves the principal equality. At odd primes, An–Wilson’s Theorem 4.4 supplies the noncyclic defect cases [9], and Späth supplies the cyclic ones [65, Proposition 6.2]. The Monster is its own required cover and has trivial outer group, so the first general argument applies at every relevant prime.

Implementation.

`ManuscriptIBAW.Sporadic.BabyNumerical` and `ManuscriptIBAW.Sporadic.MonsterNumerical` perform the subtractions on the actual Brauer and weight fibres of the complete block decomposition. The numerical sources identify the table ranks or counts of regular classes with the corresponding Brauer counts and supply the published weight totals. The equality for small defect is a separate source clause. The principal equality and

`all_block_counts` are deductions. The Baby module's `seven_transcript_values` checks the numerical data 222 and [24, 24, 21, 24]. The corresponding table calculation is described below.

The namespaces `Sporadic.BabyTwo`, `Sporadic.BabyOdd`, `Sporadic.MonsterTwo` and `Sporadic.MonsterOdd` are defined in `ManuscriptIBAW.Sporadic.BabyApplication` and `ManuscriptIBAW.Sporadic.MonsterApplication`. Each defines `model` from the selected groups, character fields, roots, blocks and covering map before defining `target` and proving `complete`. The proofs for $\ell = 2$ use the numerical deductions above. The proofs at odd primes use the published cyclic and noncyclic equalities, together with the stated local extension and block hypotheses.

The projection is bijective in the Baby case for $\ell = 2$ and both Monster cases. For the Baby Monster at odd primes it retains the actual universal double cover, with kernel of order two. Composing with the identification with the named simple group gives projection into the original simple group. That identification remains external. The character counts and reductions must agree with their ordinary complex character interpretation. This is supplied by the stated splitting or scalar comparison, for example by choosing $K = \mathbb{C}$.

5.7 The finite computations and the final theorem

The calculations for the Fischer group use `fi24blocks.g`, `fi24p3.g` and `fi24weights.g`. The calculations for the Baby Monster and Monster use `baby.g` and `monster.g`. The five table programs were checked with GAP 4.16.0 [38] and CTblLib 1.3.11 [14]. The programs `o7radical.g`, `o7weights.g` and `sp6.g` also use AtlasRep 2.1.11 [72]. The table calculations for the exceptional Type B case use `o7blocks.g`, `o7brauer.g` and `o7block2.g`. The saved outputs of all eleven programs supply the assertions and numerical results in the manuscript's computational appendix. This evidence concerns the table calculations. The radical subgroup enumerations are supplied by the cited classification and its recorded computations.

Implementation.

The accompanying computation records identify the programs, preserve their outputs and record the comparisons with the stated results. Lean checks the stated arithmetic properties of the matrices. Their identification with character values, and the interpretation of the fusions and permutations as group actions, remain explicit assumptions.

The saved Fischer data were checked in 65,088 comparisons. Of these, 23,208 compared exported values with Lean expressions. The remaining 41,880 checked consistency within the exported GAP data. The comparisons are divided as follows.

Comparison	Number
Character matrix entries against the Lean expressions	23,200
Local evaluations against the formulas in <code>Fi24Three</code>	8
Inflation values within the GAP export	408
Induced values for the 32 returned fusion candidates	20,736
Central character ratios recomputed from the export	20,736
Total	65,088

The total includes repeated entries and derived values, rather than 65,088 distinct table entries. The checks also record row and class order, root encodings and the specified permutations. The eight local evaluation comparisons concern the formulas recorded in the computation evidence. They do not establish a fusion for an automorphism of the actual local group. The prime three application uses the published fixed count with an explicit identification of the four local characters. The group, character, inclusion and coefficient interpretations remain assumptions.

The table identifications, subgroup embeddings and completeness of the radical subgroup classification remain assumptions. Block numbers refer to positions in each table. The classifi-

cation identifies the noncyclic blocks. Defect exponents zero and one establish the remaining cases.

Ordinary restriction ranks give the Brauer counts. For an invariant block whose ordinary restrictions span its Brauer space, twice the symmetrised rank minus the original rank gives the fixed count. The library manual proves these formulas and explains their finite certificates. The character value and action identifications remain assumptions.

Finally, the result for these four groups combines with the verification for the other 22 sporadic groups in the CTBlocks 0.9.5 overview [13]. That source concerns the inductive condition in Späth's Definition 4.1. It is applied to the specified covering groups and character data for those 22 groups.

Implementation.

In `ManuscriptIBAW.Sporadic.Theorem`, `BoundaryInputs` fixes the group models before their inductive conditions are proved. Its `proposition_5_4` combines the four families and the enumeration of 45 distinct pairs. `PublishedTwentyTwoInputs.published` assumes the full condition reported by CTBlocks for the other 22 groups, on the specified covers and character interpretations. The proofs reported in the overview are not formalised here. Using the division into these cases and the stated prime divisors, `Inputs.complete` proves Theorem 5.1. The group and character table identifications remain assumptions.

Chapter 6

The main theorem and the finite group consequence

Manuscript Theorem 1.1 asserts the inductive blockwise Alperin weight condition for every nonabelian finite simple group of Type B or Type C and every sporadic simple group, at each prime dividing its order. The Type B clause includes ranks one and two and groups over fields of even characteristic. The arguments of the preceding chapters establish the three cases under their stated hypotheses: manuscript Theorem 3.1, Corollary 4.15 and Theorem 5.1.

6.1 Combining the three families

Fix an identification of the simple group S with a group in one of these families. For Type C, use the specified projective symplectic group over its finite field, with the exceptional nonsimple case excluded. For Type B, use the specified orthogonal group in rank at least one. The cases in small ranks and even characteristic combine with the theorem in higher rank over odd fields. The latter uses the defining and odd nondefining cases and the arguments for $\ell = 2$. For a sporadic group, use its specified realisation and the cover attached to the chosen prime.

The theorem for each family gives the full inductive condition on the specified cover. Compose its covering projection with the chosen isomorphism to S . The covering group, characters, roots and compatible family then give the condition for S . The isomorphism preserves group order and hence divisibility by ℓ .

The Type B case thus follows from the hypotheses of the individual cases and the division into cases by rank and field. The sporadic case combines the arguments for the four groups treated in Chapter 5 with the result reported by CTBlocks for the other groups.

Implementation.

`ManuscriptIBAW.MainClassical` defines `Main.ClassicalInputs` and proves `Main.typeC_complete` and `Main.typeB_complete`. The latter uses the Type B classification of cases and the theorem in higher rank obtained from `OddHighInputs.complete`. `Main.of_allPrimeFamily` uses a `FamilyCertificate`, which contains the covering group, its projection, an isomorphism from its simple quotient to S , and a `FamilyWitness`. The classical conclusion of `Main.theorem_1_1` requires these full family data. Its sporadic alternative retains the named group, model and full inductive condition. `ManuscriptIBAW.Main` uses `Main.Presentation` and `SupportedPresentation` to prove the three cases in `Main.theorem_1_1`.

6.2 Reduction to the simple sections of a finite group

Let H be finite and let ℓ be prime. A nonabelian simple group involved in H is isomorphic to a quotient U/N , where $U \leq H$, $N \triangleleft U$, and U/N is nonabelian simple. The hypothesis of manuscript Corollary 1.2 gives an identification with a Type B, Type C or sporadic simple group for each such quotient whose order is divisible by ℓ .

If ℓ divides $|U/N|$, apply the main theorem using this identification. Otherwise, the universal ℓ' -covering group of U/N has order prime to ℓ . Every ℓ -block of that cover has defect zero, and the correspondence at $Q = 1$ satisfies the inductive condition, as in the proof of manuscript Corollary 1.2.

Späth's reduction theorem then gives the blockwise Alperin weight equality for H [65, Theorem A]:

$$|\mathrm{IBr}(b)| = |\mathrm{Alp}(b)| \quad \text{for every block } b.$$

Here $\mathrm{IBr}(b)$ consists of the irreducible Brauer characters in b , and $\mathrm{Alp}(b)$ consists of the conjugacy classes of weights belonging to b , with their ordinary local characters of defect zero. The specified local block operations, character realisations and root choices describe these objects.

To apply the reduction, each complete family must satisfy the inductive condition in that theorem. In the exceptional Type B case, an explicit transport assumption first interprets the blockwise criteria on the natural quotients as relative block conditions on the common family over $3.\Omega_7(3)$. This includes character and block identifications over the fixed coefficient fields, with compatible root correspondences for quotient groups, extension groups and intermediate subgroups. A separate published passage combines those relative block conditions into a full family [50, Lemma 3.3], with the normalisation of Späth's Definition 5.17 and Remark 5.18 [65]. This passage chooses compatible global correspondences when necessary. The proof of Koshitani–Späth's lemma also supplies the abstract lifting of characters, weights and blocks from faithful central quotients. Interpreting these objects in the fixed common family, with the coefficient and root identifications described above, remains the separate assumption `NaturalQuotientTransport`.

The formal proof applies the published reduction under these identifications and the defect zero implication for sections of order prime to ℓ . The general reduction theorem is an explicit external hypothesis. The deduction supplies its hypotheses for every nonabelian simple section of H .

Implementation.

`Main.corollary_1_2` constructs `SectionEvidence.supported` from the main theorem when ℓ divides the section order and `SectionEvidence.primeTo` otherwise. It applies `Main.FiniteGroupReductionSource.reduce`. The source is indexed by the block family, its character and root data, and `SpathCoefficientField`. This requires the algebraically closed modular field k to be algebraic over its prime field $\mathbb{F}_\ell$. Its antecedent uses the local `Main.SectionEvidence`, which requires the full classical or sporadic conclusion on each actual simple section, or that its order is prime to the chosen prime. The external assumption combines the published reduction with the defect zero implication for sections of order prime to ℓ . Its conclusion is `BlockwiseAWC` for that family.

Appendix A

Manuscript statements and Lean declarations

This table gives an entry point for each numbered manuscript statement. The selected declarations include definitions, deductions and explicit assumptions. Their association with a statement does not assert that every part of its proof is formalised or that its external hypotheses have been established. The relevant chapters explain those hypotheses and the full types in Lean specify them.

The source associations accompanying the package list the additional declarations used by each argument. Numbers within declaration names are not manuscript cross-references.

Manuscript	Selected declaration	Role and scope
Theorem 1.1	<code>ManuscriptIBAW.Main.theorem_1_1</code>	Conditional deduction for the three families.
Corollary 1.2	<code>ManuscriptIBAW.Main.corollary_1_2</code>	Deduction using the assumed finite group reduction.
Definition 2.1	<code>ManuscriptIBAW.Preliminaries.CentralSectors.existsUnique_centralCharacterSector</code>	Existence and uniqueness of the central character sector.
Lemma 2.2	<code>ModularRep.LinearBlockStabilizer.modular_block_stabilizer_bound</code>	Modular stabiliser bound under the block and character hypotheses.
Theorem 2.3	<code>ManuscriptIBAW.Preliminaries.SpathCriterion.current_theorem_2_3</code>	Character triple equivalence under an explicit source assumption.
Remark 2.4	<code>ManuscriptIBAW.Preliminaries.BlockFamilies.fullFamily</code>	Part (a): construction of the family using the assumed passage from block conditions to the full condition. Part (b) is an orbit transport observation.
Corollary 2.5	<code>ManuscriptIBAW.Preliminaries.Descent.current_corollary_2_5_coherent</code>	Descent with explicit character and root compatibility.
Lemma 2.6	<code>ManuscriptIBAW.Preliminaries.CoreCoreLeRaw</code>	Normal core containment in the central quotient construction.
Corollary 2.7	<code>ModularRep.PaperProofs.ConlonBasicSet.corollary_2_4_conlonMark</code>	Conlon deduction for the stated prime and lattice assumptions.
Lemma 2.8	<code>ManuscriptIBAW.equivariant_bijection_of_basic_set_fixed</code>	Equivariant bijection from a fixed integral basic set.

Manuscript	Selected declaration	Role and scope
Lemma 2.9	<code>ManuscriptIBAW.Jordan.SemidirectExtension.restrict</code>	Restriction of an extension from a semidirect product.
Lemma 2.10	<code>ManuscriptIBAW.TypeC.EvenApplication.unipotent_bawGood</code>	Application of the cyclic outer criterion under its source assumptions.
Lemma 2.11	<code>ManuscriptIBAW.FiniteCharacterSelection.exists_delta_injection_of_nonnegative</code>	Finite character selection. Geometry and trace identities remain inputs.
Lemma 2.12	<code>ManuscriptIBAW.TypeC.GGGRSelection.delta_selection</code>	Conditional Type C selection from the specified geometric sources.
Theorem 3.1	<code>ManuscriptIBAW.TypeC.all_primes</code>	Conditional deduction for the Type C cases.
Lemma 3.2	<code>ManuscriptIBAW.Jordan.generalPerLabel_of_fullActor</code>	Restriction of the Brauer character hypothesis to the automorphism groups chosen for individual semisimple parameters.
Proposition 3.3	<code>ManuscriptIBAW.TypeC.prop33_originalIBAW</code>	Principal block application with explicit field, counting and compatibility inputs.
Remark 3.4	<code>ModularRep.PaperProofs.OddTwoWreathCoreCharacterSource.triangular_ne_two</code>	Arithmetic distinction only. The subgroup identifications remain external.
Proposition 3.5	<code>ManuscriptIBAW.TypeC.prop34_full_block_condition</code>	Conditional rank two conclusion under the sources for the selected prime and field.
Lemma 3.6	<code>ModularRep.PaperProofs.EvenFieldLemmas35_36Actual.lemma_3_5_actual</code>	Field invariance of the specified ordinary characters and generic weights under the source assumptions.
Lemma 3.7	<code>ModularRep.PaperProofs.EvenFieldLemmas35_36Actual.lemma_3_6_actual</code>	Equivariant correspondence between the ordinary character set and generic weights under the source assumptions.
Proposition 3.8	<code>ManuscriptIBAW.TypeC.EvenApplication.unipotent_definition35</code>	Unipotent block condition under the specified published criterion.
Proposition 3.9	<code>ManuscriptIBAW.TypeC.EvenApplicationRouting.fullHG_strictBlocks_of_proposition39_semisimple</code>	Jordan reduction from the semisimple parameter hypotheses.
Lemma 3.10	<code>ManuscriptIBAW.quotientKernelCenterEquiv</code>	Quotient and kernel comparison used in the stabiliser bound.
Proposition 3.11	<code>ManuscriptIBAW.TypeC.OddConformalCorrespondence.globalCorrespondence</code>	Conformal correspondence under the source and action assumptions.
Lemma 3.12	<code>ManuscriptIBAW.TypeC.OddGlobalFactorization.GlobalBijection.allBrauerFactorization</code>	Transfer of ordinary stabiliser factorisation to Brauer characters.
Remark 3.13	<code>ManuscriptIBAW.TypeC.OddGlobalFactorization.BlockSourceInputs</code>	Record of the assumptions used for the Conlon argument. It supplies neither canonical labels nor unitriangularity.

Manuscript	Selected declaration	Role and scope
Proposition 3.14	<code>ManuscriptIBAW.TypeC.OddPrimeApplication.full_block_condition_source_instantiated</code>	Application of the full criterion under the listed sources.
Theorem 4.1	<code>ManuscriptIBAW.TypeB.AllPrimes.all_primes</code>	Conditional deduction for the Type B cases.
Lemma 4.2	<code>ManuscriptIBAW.TypeB.OddConlon.block_bijection</code>	Blockwise Conlon correspondence under the stated basic set hypotheses.
Proposition 4.3	<code>ManuscriptIBAW.TypeB.OddUniform.complete</code>	Odd prime application with its structural and criterion sources.
Lemma 4.4	<code>ManuscriptIBAW.TypeB.Levi.component_return_full</code>	Action on a factor after a cycle of the factor permutation.
Lemma 4.5	<code>ManuscriptIBAW.TypeB.Levi.cartesian_character_orbit</code>	Character orbit calculation for the rational product.
Lemma 4.6	<code>ManuscriptIBAW.TypeB.Levi.lemma46</code>	Clifford stabiliser deduction on the specified characters.
Lemma 4.7	<code>ManuscriptIBAW.TypeB.PrincipalField.rationalClass_fixed</code>	Field invariance of the specified rational classes.
Lemma 4.8	<code>ManuscriptIBAW.TypeB.PrincipalSeries.lemma_4_8</code>	Principal series deduction under its source interpretation.
Proposition 4.9	<code>ManuscriptIBAW.TypeB.GGGRBasis.gggr_basis</code>	Conditional basis of projected generalised Gelfand–Graev characters from the geometric and finite selection inputs.
Corollary 4.10	<code>ManuscriptIBAW.TypeB.PrincipalApplication.principal_selector</code>	Principal character selection using field invariance and extension inputs.
Proposition 4.11	<code>ManuscriptIBAW.TypeB.Exceptional.Inputs.complete</code>	Exceptional cover application with computation, quotient and root assumptions.
Lemma 4.12	<code>ManuscriptIBAW.Jordan.TypeBLocalized.full_family_of_hypotheses</code>	Localised Jordan reduction under its factor and transport hypotheses.
Proposition 4.13	<code>ManuscriptIBAW.TypeB.TwoApplication.Inputs.brauerHypothesis</code>	Derivation of the required Brauer character hypothesis.
Proposition 4.14	<code>ManuscriptIBAW.TypeB.TwoApplication.Inputs.complete</code>	Prime two application under the principal and Levi sources.
Corollary 4.15	<code>ManuscriptIBAW.TypeB.finite_simple_typeB</code>	Conditional deduction for the specified Type B groups.
Theorem 5.1	<code>ManuscriptIBAW.Sporadic.Theorem.Inputs.complete</code>	Sporadic deduction including the external result for 22 groups.
Lemma 5.2	<code>ManuscriptIBAW.Sporadic.CompleteCollapse.lemma_5_2</code>	Numerical construction with compatible roots for the constructed family.
Lemma 5.3	<code>ManuscriptIBAW.Sporadic.Cancellation.cancel_equivariant_equiv</code>	Equivariant cancellation of the specified parts.
Proposition 5.4	<code>ManuscriptIBAW.Sporadic.Theorem.BoundaryInputs.proposition_5_4</code>	Four remaining groups under their numerical and compatible packet sources.

Appendix B

Sources used in the manuscript

This index groups the manuscript citations by reference and result locator. The manuscript line numbers locate each occurrence in the accompanying manuscript. Counts describe citation occurrences. They do not certify the application of a source result or imply that it has been proved in Lean.

Reference and source locator	Manuscript lines	Occurrences
[1]	116	1
[2, (6D)]	953, 1153	2
[2, p. 190]	174, 743	2
[2, p. 198]	967	1
[3]	158	1
[4]	3752	1
[4, Lemma 4.2(c) and Theorem 4.3]	3389	1
[4, Lemma 4.2(d)]	3332, 3346	2
[4, Proposition 4.1 and Table 1]	3386, 3421, 3430, 3765, 3802	5
[4, Table 2]	3344	1
[4, Theorem 4.3]	3264	1
[5]	3317	1
[5, Lemma 4.7]	3306	1
[5, Section 4.3.4]	3415, 3448	2
[5, Table 8]	3339, 3343, 3348, 3356, 3370, 3380, 3410, 3448	8
[5, proof of Theorem 4.2 and Table 17]	3516	1
[5, proof of Theorem 4.2 and Table 2]	3483	1
[6]	3484, 3517	2
[7, Theorem 5.2]	3257	1
[8, Lemma 5.2]	3495, 3834	2
[8, Theorem 5.3]	3499, 3501, 3836	3
[9, Theorem 4.4]	3528	1
[10, Example 4.8]	1535	1
[10, Proposition 2.3(b)]	2566	1
[10, Proposition 4.11(a)]	1106, 4320	2
[10, Proposition 5.3(a)]	2313, 2568	2
[11, Lemma 7.4]	2899	1
[12, Theorem 2.3.2 and Section 3.2]	507	1
[12, Theorem 3.5.5]	517	1
[13]	200, 3168, 3542	3
[14]	2769, 3391, 3569	3
[15, Section 7]	3479, 3824	2

Sources used in the manuscript

Reference and source locator	Manuscript lines	Occurrences
[16, Section 5]	3512, 3840	2
[17, Section 2]	777	1
[17, Theorem 2.2]	781	1
[18, Theorem 1.1]	1173	1
[18, p. 1203]	2945	1
[19, Remark 4.4]	399	1
[19, Theorem 4.5]	1731, 1747, 1792, 1796, 1815, 1817	6
[19, Theorem 4.5(ii)]	1783	1
[20, Theorem 4.4(i)]	1226, 1399	2
[20, Theorem 4.4(iii)]	1222	1
[21, Proposition 15.6(i)]	1679, 1960	2
[21, Proposition 15.7]	1244	1
[21, Proposition 9.8(iv)]	1032, 2575	2
[21, Theorem 21.14]	1034, 1109, 2310, 2657	4
[21, Theorem 9.12]	2884	1
[22, Section 2.3]	1259	1
[22, Theorem 2.8]	1395	1
[22, Theorem 3.1]	1621	1
[22, p. 700]	1229, 1310	2
[23, Theorem 3.1]	1694	1
[24, Proposition 2.7]	2593, 2655	2
[24, Proposition 2.8]	987	1
[24, Theorem 2.9]	985, 1038	2
[25]	148, 516	2
[26, Table 1]	3237	1
[27, Theorem 8.1]	2935, 2968, 3016	3
[28, Proposition 7.2]	2013	1
[28, Proposition 7.5]	2037	1
[28, Remark 2.2]	2018	1
[28, Section 3.1]	1909	1
[28, Theorem 1]	143, 2004, 2044	3
[28, Theorem 1 and Assumption 3.11]	1902	1
[28, Theorem 2.1]	2000	1
[28, Theorem 2.3]	1954	1
[28, Theorem 3.10]	2026	1
[28, proof of Lemma 5.16]	1980	1
[29]	832	1
[29, Proposition 4.6]	849, 3151	2
[30, Corollary 2.13]	2678, 2755, 3177, 3194, 3470	5
[30, Lemma 2.1]	459	1
[30, Lemma 2.3(iii)]	3299	1
[30, Lemma 2.5]	2716, 3303	2
[30, Section 2.3]	778	1
[30, Theorems 1.2 and 1.4]	140	1
[30, Theorem 1.5]	133	1
[31]	170, 180	2
[31, Assumption 5.3]	824, 2804, 3122	3
[31, Definition 3.5]	344	1
[31, Hypothesis 5.5(a)]	825	1
[31, Hypothesis 5.5(b)]	826	1
[31, Lemma 5.1 and the proof of Proposition 5.2]	3064	1
[31, Proposition 5.6]	2928	1
[31, Remark 5.8]	768	1

Reference and source locator	Manuscript lines	Occurrences
[31, Remark 5.9]	1422	1
[31, Section 3.5]	476	1
[31, Section 4.3]	815	1
[31, Section 5]	798, 804, 822	3
[31, Theorems 3.18 and 5.7]	482	1
[31, Theorem 3.18]	575, 591	2
[31, Theorem 5.7]	827, 2890, 2926	3
[32]	832	1
[32, Theorem 1]	838	1
[32, p. 6518]	203	1
[33, Corollary 4.3]	983, 1049	2
[33, Corollary 4.6]	1081, 1123	2
[33, Lemma 5.1]	1044	1
[33, Proposition 3.4]	1062, 1122	2
[33, Proposition 5.4]	944, 954, 975, 1154	4
[33, Proposition 5.5]	1047	1
[33, Proposition 5.9]	1048	1
[33, Proposition 6.1 and the proof of Theorem 6.2]	1050	1
[33, Remark 3.5]	1124	1
[33, Sections 2 and 5]	740	1
[33, Section 4.1]	991, 1041	2
[33, Section 5]	766, 901	2
[33, Section 5.1]	937	1
[33, Theorem 1]	173, 872	2
[33, p. 11]	747	1
[34]	163, 1381	2
[34, Condition 3.23]	1211	1
[34, Condition 6.1]	1432, 1436	2
[34, Definitions 3.17–3.18]	1203	1
[34, Definition 3.5]	1198	1
[34, Lemma 3.21]	1413	1
[34, Proposition 3.20]	1412	1
[34, Proposition 3.24(b)]	1214	1
[34, Section 3.2]	1201	1
[34, Theorem 6.2]	1450	1
[34, Theorem 7.5]	1402	1
[35, Proposition 7.3]	1577	1
[36]	742	1
[36, Corollary 2.22(2)]	2699, 3686	2
[36, Corollary 5.15]	2640	1
[36, Corollary 5.17]	2660, 2667	2
[36, Lemma 2.3]	904, 973, 1149, 3685	4
[36, Lemma 5.7 and the proof of Proposition 5.11]	2687	1
[36, Proposition 3.48]	935, 947	2
[36, Proposition 3.48(4)]	913	1
[36, Proposition 5.11]	2690, 2696	2
[36, Question 5.27]	2650	1
[36, Remark 3.36]	177	1
[36, Remark 3.47(2)]	1131	1
[36, Remark 3.49]	749, 758	2
[36, Remark 3.50]	762, 911, 934, 1130	4
[36, Section 3]	1138	1
[36, Theorem 1]	2793	1
[36, Theorem 1 and Assumption 5.1]	184	1
[36, Theorem 3.37]	906	1

Reference and source locator	Manuscript lines	Occurrences
[36, Theorem 5.16]	2643	1
[36, proof of Proposition 3.48]	746, 757, 950	3
[37]	158	1
[38]	3568	1
[39]	162	1
[39, Proposition 2.2]	2493	1
[39, Theorem A]	1440, 1671	2
[40, Corollary 3.6(b)]	3891, 3930	2
[40, Lemma 3.5]	3890, 3919, 3923, 4067	4
[40, Lemma 3.5 and Corollary 3.6(b)]	4050	1
[41]	3966	1
[41, (2.3)(a)]	4028	1
[41, (2.4)]	3963, 4016, 4044, 4399	4
[41, (2.4)(a)]	4074	1
[41, (2.4)(b)]	4078	1
[41, (2.4)(c)]	4077	1
[41, (3.3)]	3976	1
[41, (3.3)(b)]	3965, 4082, 4139, 4401	4
[41, (4.2)]	3912	1
[41, (4.3)]	3968, 4183, 4424	3
[41, (4.4)(a)]	4196, 4354	2
[41, (4.4)(a)–(b)]	4193, 4333, 4433	3
[41, (4.4)(b)]	4197, 4369	2
[41, (4.7)]	4004	1
[41, (4.7)(a)–(c)]	3994, 4384	2
[41, (4.9)]	3972, 4375	2
[41, (6.1)(c)]	4246, 4479	2
[41, Basic assumptions, p. 140]	3967	1
[41, Corollary 3.2]	4142, 4171, 4416	3
[41, Propositions 5.3 and 5.5]	3974, 4386	2
[41, Propositions 5.3 and 5.5]	4390	1
[41, Proposition 3.1]	4167, 4412	2
[41, Proposition 5.3]	3997	1
[41, Remark 3.4 and the proof of Theorem 3.8]	3983, 4380	2
[41, Theorem 3.8]	3991, 4173	2
[41, Theorem 4.5]	3959, 3962, 4017, 4191, 4198, 4204, 4371, 4434, 4445	9
[41, Theorem 7.1]	4187, 4429	2
[41, p. 151]	3988	1
[42, Corollary 3.5]	4218, 4464	2
[42, Proposition 2.3]	4198	1
[42, Proposition 2.4]	2367	1
[42, Proposition 4.3]	988, 1015, 3858	3
[42, p. 821]	2403	1
[43, (1.1)–(1.2)]	537	1
[43, Theorem 5.1]	1633	1
[44, Section 3(A)]	3917, 4456	2
[44, Section 3(B), p. 443]	630	1
[44, Section 3(C)]	4242	1
[44, Section 3(C) and Theorem 3.7]	4475	1
[44, Theorem 3.7]	4244	1
[45, 4.1.1]	2326	1
[45, Definition 4.1.17]	648	1
[45, Example 2.7.18]	1027	1

Reference and source locator	Manuscript lines	Occurrences
[45, Lemma 1.5.22]	4322	1
[45, Proposition 2.6.16]	1030	1
[45, Proposition 3.4.21]	2434, 4151, 4247, 4480	4
[45, Proposition 3.4.3]	1018	1
[45, Proposition 3.4.3 and Corollary 3.4.5]	2467	1
[45, Remark 1.5.12]	2190	1
[45, Remark 2.3.28]	632	1
[45, Remark 4.2.1]	1314	1
[45, Remark 4.2.17]	2409, 4352	2
[45, Theorem 1.7.15]	2458, 3035	2
[45, Theorem 4.7.1(3)]	1970	1
[46, Theorem 8.2]	1472	1
[46, Theorem 8.2 and Corollary 8.7]	1572	1
[47, Corollary 11.22]	570	1
[48, Theorem 1.1]	2723	1
[49, Definition 2.1 and Remark 2.2]	1398	1
[49, Theorem A(e)]	1400	1
[50, Definition 3.2]	349	1
[50, Lemma 3.3]	403	1
[50, Theorem 1.1]	2045	1
[50, Theorem 1.3]	2737	1
[51, (6.2.11)]	3905, 4014, 4084, 4097, 4108	5
[51, Assumption 5.0.14]	3901	1
[51, Remark 4.4.6]	4100	1
[51, p. 115]	4007, 4389	2
[52, Lemmas 5.1 and 5.3]	1757	1
[52, Lemma 3.13]	1749	1
[52, Lemma 5.1]	1760	1
[52, Lemma 5.3]	1763	1
[52, Lemma 5.4]	1721	1
[52, Lemma 5.5]	1800	1
[52, Lemma 5.5(1)–(2)]	1801	1
[52, Remark 2.2(1)]	1790	1
[52, Remark 2.5]	1620	1
[52, Section 2.C]	1799	1
[52, Section 6]	156	1
[52, Theorem 2]	142, 1180, 1571, 1716	4
[52, Theorem 3]	160, 1469, 1471	3
[52, Theorem 4.13 and Remarks 4.7–4.8]	1750	1
[53, Theorem 1.1]	883, 1171	2
[54, (8.4)(a)]	4044, 4133	2
[54, Proposition 7.2]	4117	1
[54, Proposition 8.5 and Corollary 8.6]	3907, 4048	2
[54, Theorem 10.7]	4184, 4425	2
[54, Theorem 11.2(iv)]	4291	1
[55, Section 5.5]	2744	1
[56, Proposition 12.14]	2155	1
[56, Proposition 24.21 and Theorem 24.24]	1895	1
[56, Remark 24.19 and Table 24.3]	847	1
[56, Tables 24.2–24.3 and Remark 24.19]	716, 1863	2
[56, Theorem 21.11]	2288	1
[56, Theorem 22.5]	3001	1
[57, Definition 3.4]	341	1
[57, Lemma 3.14]	471	1

Reference and source locator	Manuscript lines	Occurrences
[57, Lemma 3.6(i)]	862	1
[57, Lemma 6.3]	437	1
[57, Proposition 8.5]	869	1
[58, Corollary 2.10]	3515	1
[58, Lemma 6.1]	3399	1
[58, Theorem 4.14]	3338, 3397	2
[58, Theorem 4.14 and Lemma 6.1]	3794	1
[58, Theorem 8.11]	2636, 2658	2
[58, Theorem 8.12]	571	1
[58, Theorem 9.10]	452, 2631	2
[58, Theorem 9.9(c)]	3281	1
[59, Lemma 2.3(c)]	3274	1
[60, Corollary 4.2]	3111	1
[60, Corollary 4.2 and Lemmas 4.3 and 4.5]	1079, 1506	2
[60, Lemma 4.5(a)–(b)]	2902, 3115	2
[61, Theorem 4.1]	2729, 2745	2
[62, Theorem 13.7]	3334, 3382, 3487, 3520	4
[62, Theorem 8.1]	2727	1
[63, Theorem 5.5]	1175, 1543, 1550, 1835, 3663	5
[63, proof of Theorem 5.5]	3638	1
[64, Theorem 3.2]	4186, 4225, 4428, 4459	4
[65, Corollary 6.3]	1553	1
[65, Definition 2.1]	295	1
[65, Definition 4.1]	118, 268, 368, 3169	4
[65, Definition 4.1(i)–(iv)]	321	1
[65, Proposition 4.6(a)–(c)]	430	1
[65, Proposition 6.2]	3259, 3444, 3450, 3460, 3467, 3502, 3529	7
[65, Remark 5.18]	404	1
[65, Theorem A]	208, 3554	2
[65, Theorem C]	737, 1169, 1829, 3132	4
[65, p. 184]	271	1
[66, Proposition 3.6(b) and Theorem 3.5(b)]	433	1
[66, Theorem 4.4]	355	1
[67, Theorem 3.3]	1353	1
[68, (3.2)]	2427	1
[68, Definition 2.1]	1003	1
[68, Proposition 2.4]	4446	1
[68, Proposition 5.4]	989, 1013, 1022	3
[68, Section 2]	2270, 2285, 2356	3
[68, Theorem 1.1]	4147, 4404	2
[68, Theorem 3.1]	2431, 2530	2
[68, Theorem 3.2]	1010, 1019, 2422, 2428, 2529	5
[68, pp. 45–46]	2507, 2517	2
[68, proof of Proposition 2.4]	1001, 4350	2
[69, 10.1–10.3]	681	1
[69, 10.3]	1009, 4335	2
[69, 6.19]	4317	1
[69, 7.1 and Proposition 10.2]	4348	1
[69, 7.4]	4357	1
[69, Definition 2.7]	4359	1
[69, Proposition 10.2]	4342	1
[69, Theorem 2.11]	4315, 4356	2

Reference and source locator	Manuscript lines	Occurrences
[69, p. 476]	2329, 4240, 4340	3
[70, (11.15)]	2446, 3885, 4027, 4263, 4493	5
[70, (11.15), Lemma 14.15 and Proposition 15.4]	4152	1
[70, 10.1]	3921	1
[70, 10.1–10.8]	3872, 4022	2
[70, 10.8]	3878	1
[70, 11.12]	3877	1
[70, 6.2]	3902	1
[70, Corollary 13.6]	3908, 4049, 4185, 4293, 4426	5
[70, Definition 6.1 and Lemma 6.3(i)]	3899	1
[70, Lemmas 14.12 and 14.15]	2561	1
[70, Lemma 10.9, (10.11), Theorem 10.14 and (10.20)]	3920	1
[70, Lemma 11.3]	4086, 4109, 4119	3
[70, Lemma 14.12]	2473	1
[70, Lemma 14.15]	2444, 2581, 4251, 4490	4
[70, Lemma 14.15 and Proposition 15.2]	1036	1
[70, Lemma 6.3(i)]	4011, 4382	2
[70, Proposition 11.5]	4038, 4122	2
[70, Proposition 15.2]	2583	1
[70, Proposition 15.4]	2445, 4261, 4491	3
[70, Proposition 4.6 and Section 5]	3880	1
[70, Theorem 11.13 and Lemma 11.16]	3896, 4046, 4393	3
[70, proof of Lemma 14.12]	1017	1
[71, Proposition 11.10]	2272, 2294	2
[71, Proposition 7.2]	1681, 1963, 3079	3
[72]	3569	1

Appendix C

Further constructions in the source library

The preceding chapters follow the manuscript arguments. This appendix records related auxiliary constructions in the source library. Their hypotheses concern the same characters, groups, coefficient fields and actions as their conclusions. They do not provide instances of the full application hypotheses.

C.1 Central kernels and local ordinary characters

Let X be a finite perfect group, let ℓ be prime, and suppose that $\ell \nmid |Z(X)|$. Fix an algebraically closed modular field k of characteristic ℓ , an ordinary field K of characteristic zero and a root correspondence ι . For a representation ρ_φ affording an irreducible Brauer character φ , put

$$Z_\varphi = Z(X) \cap \ker \rho_\varphi, \quad \overline{X}_\varphi = X/Z_\varphi.$$

Factoring ρ_φ through this quotient preserves irreducibility. With the induced root convention, inflation recovers φ . Perfectness shows that the centre of the quotient is the image of $Z(X)$: commutators with an element central modulo Z_φ define a homomorphism from X to the central subgroup Z_φ , and that homomorphism is trivial. The quotient character is therefore faithful on the quotient centre.

For a weight (Q, θ) in the block of φ , the local reduction and block compatibility hypotheses identify the scalar character of its local Brauer character with that of φ on $Z(X)$. Consequently Z_φ is also in the local modular kernel. Root agreement on these central elements transfers this equality to the ordinary character θ . Under the specified normaliser quotient and radicality hypotheses, ordinary descent constructs $\bar{\theta}$ with

$$\theta = \bar{\theta} \circ q_Q, \quad q_Q : N_X(Q)/Q \longrightarrow N_{\overline{X}_\varphi}(\overline{Q})/\overline{Q}.$$

The map q_Q is the map induced by this central quotient. Its surjectivity gives uniqueness of $\bar{\theta}$. Block descent additionally uses the complete quotient block decompositions and preservation of the relevant primitive idempotents; the character factorisation alone does not supply those facts.

Implementation.

The modules

`ModularRep.PaperProofs.SporadicFi24P3Definition44NamedCarrierFullCenterScalar` and `ModularRep.PaperProofs.SporadicFi24P3Definition44NamedCarrierOwnCentralQuotient` contain the scalar and quotient constructions. The former proves `full_center_scalar_eq`; the latter constructs `ownCentralQuotientBrauerSource`. The hypotheses on local block operations, quotient normalisers and coefficient conventions are retained.

An equivariant block-preserving bijection $\Omega : \text{IBr}_\ell(X) \simeq \mathcal{W}(X)$ also determines ordinary local maps. For each radical subgroup Q , take the Brauer fibre

$$\{\varphi : \text{rad}(\Omega\varphi) = [Q]\}.$$

The map $\theta \mapsto [(Q, \theta)]$ identifies the defect zero characters of $N_X(Q)/Q$ with weight classes whose radical subgroup has conjugacy class $[Q]$. Its inverse defines Ω_Q by

$$[(Q, \Omega_Q(\varphi))] = \Omega(\varphi).$$

Injectivity gives the covariance of these local maps under automorphisms stabilising Q . The partition retains empty fibres. Its nonempty parts are distinguished by their radical classes; equality of two empty parts does not identify their labels.

C.2 Extensions and compatible roots

The local extension construction for φ uses a finite ambient group A with a normal subgroup $B \cong \overline{X}_\varphi$ and a map $f : X \rightarrow A$ with kernel Z_φ . For $D = N_A(f(Q))$, the specified local map $f_N : N_X(Q) \rightarrow D$ satisfies

$$\iota_D f_N = f \iota_{N_X(Q)}, \quad \text{im}(f_N) = B \cap D.$$

The ambient hypotheses include $C_A(B) = Z(A)$ and $\ell \nmid |Z(A)|$. The quotient $A/Z(A)$ identifies with the stabiliser of the original character, using the opposite automorphism group to implement the manuscript's right action. The witnesses include global and local Brauer extensions, their irreducible restrictions for every $B \leq J \leq A$, and the required block induction relations.

The raw witness permits separate root correspondences. The module `ModularRep.PaperProofs.SpathRootCoherence` adds the agreement conditions needed by the full inductive certificate. For `OriginalPacketRoots`, the original and quotient conventions agree on the quotient roots; the global extension convention agrees with the quotient and local extension conventions, and with the conventions on J and $D \cap J$ for every intermediate J . When the original group embeds in the extension group, `ActualPacketRoots` additionally requires agreement with its original roots. Factoring through the central kernel is what permits the first construction to use the weaker condition.

Let H be a finite group and $N \triangleleft H$ with $[H : N] \leq 2$. A finite dimensional irreducible representation ρ of N over an algebraically closed field, invariant under conjugation by H , extends to H on the same vector space. In the index two case, choose $t \in H \setminus N$ and an invertible intertwiner T . Schur's lemma makes $\rho(t^2)^{-1}T^2$ a nonzero scalar c . Rescaling T by a square root of c^{-1} gives an operator U satisfying $U^2 = \rho(t^2)$; together with ρ , this defines the representation on both cosets. The construction also applies to a nonsplit extension. It is implemented in `ModularRep.PaperProofs.SporadicFi24P3Definition44NamedCarrierIndexTwoExtension`.

These extension constructions do not turn a numerical bijection into the full inductive condition without the remaining block and root hypotheses. In the triple-cover applications, compatible replacement witnesses are supplied for the same numerical correspondence, as described in the sporadic chapter. Their existence remains an assumption.

C.3 Selected Type B constructions

The rank-three Morita construction keeps one Levi character, its representation, its actual field stabiliser and its chosen Jordan image throughout. The base functor is the balanced tensor product with the specified block bimodule. The upper functor acts on the extensions by the same field stabiliser. Their restriction square gives

$$\mathrm{Res}_G \hat{\mathcal{F}} \cong \mathcal{F} \mathrm{Res}_L .$$

Applying the upper functor transfers a Levi extension to the ambient group. Conversely, essential surjectivity of the upper equivalence and full faithfulness of the base equivalence recover an extension of the original Levi representation. The stated equality of tensor characters with the Jordan map identifies the resulting character. These are hypotheses and deductions for the selected bimodule and actions, rather than an interpretation of an arbitrary abstract Morita equivalence.

Implementation.

The selected result is `ModularRep.PaperProofs.TypeBRankThreeMoritaSelectedApplication.actual_selected_morita_extension`. The separate module `ModularRep.PaperProofs.TypeBRankThreeSameFieldAutomorphism` retains the identification between the relevant outer action and the normalised field action on this same union of blocks. The field stabiliser is not identified with a full automorphism stabiliser by the Morita equivalence alone.

For a finite direct product $G = \prod_i G_i$ of finite groups, a radical 2-subgroup R is the product of its coordinate images R_i . The argument uses radicality, the normaliser condition for finite 2-groups and the product formula for the 2-core. The actual coordinate maps then give

$$N_G(R)/R \cong \prod_i N_{G_i}(R_i)/R_i.$$

The supplied ordinary tensor realisations, their unique factorisation and the defect zero criterion identify the defect zero characters on these quotients. Passing to conjugacy classes gives the product correspondence on weights. To retain blocks, the proof separately uses the ordinary block product identities and checks the central character equation along the product normaliser inclusion. The resulting `ModularRep.PaperProofs.TypeBRankThreeProductRawWeightBlock.rawWeightBlock_primitive` therefore identifies the ambient primitive block induced from the actual ordinary local blocks.

Implementation.

The three Type B modules named in this section are not imported by any other module in the current formalisation. They record auxiliary constructions; this section does not assert their integration into the manuscript deductions.

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